



DD2380 Artificial Intelligence

Hidden Markov Models (HMMs)

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Lectures

- Intro and practicalities
- Ethics
- Problem Solving
 - Search
 - Adversarial Search
- **Taming Uncertainty**
 - Probabilistic Reasoning
 - **Hidden Markov Models (HMMs)**
- Invited talk – Multi Agent Systems + Robotics
- Invited talk – Computer Vision
- Making decisions under uncertainty
 - Decision Making
 - Reinforcement Learning (RL)
- Invited talk – Machine Learning (ML)
- Invited talk – Natural Language Processing (NLP) and Conversational AI
- Knowledge, Reasoning, and planning
- Invited talk – Generative AI & Final QA

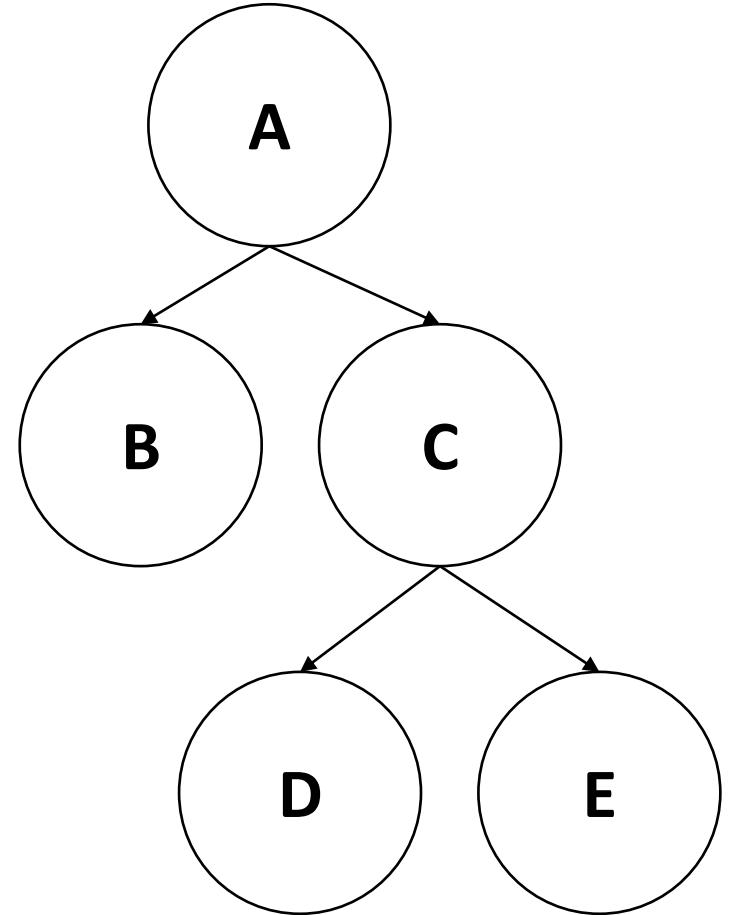


Reading Instructions

- Chapter 15 (3rd edition) or Chapter 14 (4th edition), Russell & Norvig.

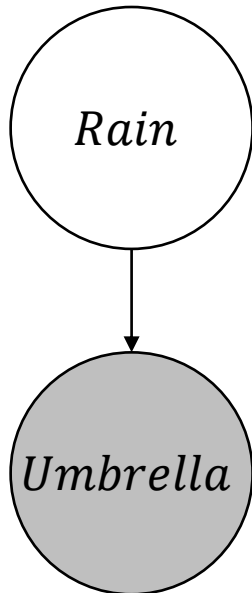
Bayesian Networks

- Aka Probabilistic **Directed Acyclic** Graphical Model
- Represents joint probability distribution (in a compact manner)
 - Helps with analyzing probability information
 - Helps with structuring probability information
- Arrow → “direct influence over”
 - A has direct influence over B
- Each arrow is accompanied with a conditional probability distribution, e.g., $P(B|A)$
- Very hard to gather data to build a model for $P(A, B, C, D, E)$, much easier to look at conditional probabilities such as $P(B|A)$
- Factorization
 - $P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$



Example: Umbrella world

- A person tries to infer $rain = \{true, false\}$ by observing if a certain person has an $umbrella = \{true, false\}$ that day.
- Draw Bayesian network!
 - What are the variables?
 - Which cause which?

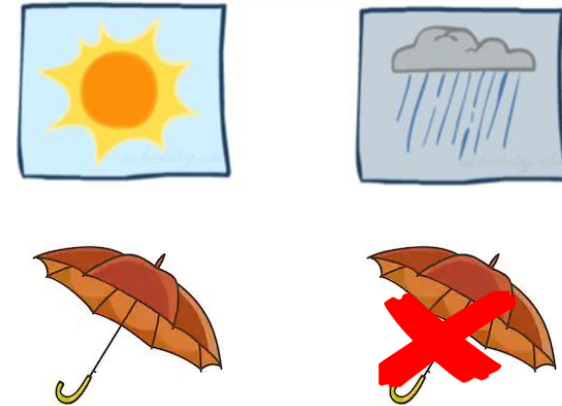


We cannot directly observe if it rains.

This is **hidden** to us.

We observe umbrella and infer if it rains or not

Rain causes the person to use an umbrella.

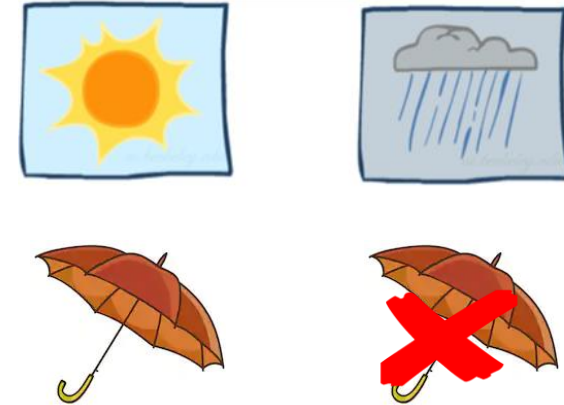
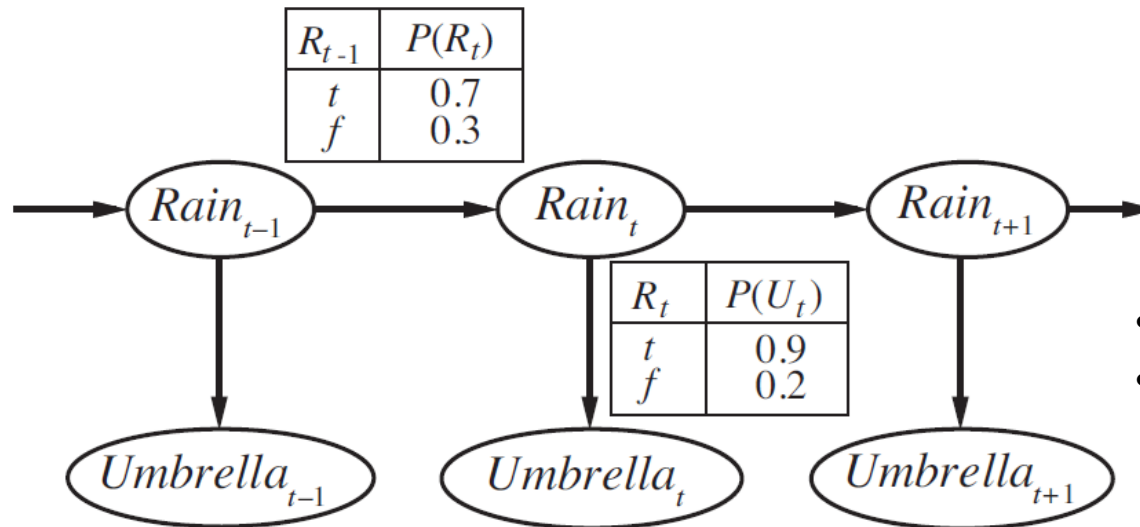


We often cannot directly observe the state!

- Other examples:
 - Cannot observe the weather, only the temperature
 - State = weather, observation = temperature
 - Cannot observe the words spoken, only the sound uttered
 - State = word, observation = sound
 - Cannot observe the position of the car only the laser scanner readings
 - State = position, observation = laser data

Example: Sequential Umbrella world

- A person tries to infer $rain = \{true, false\}$ **every day** by observing if a certain person has an *umbrella* = $\{true, false\}$ over consecutive days.
- Draw the Bayesian network that corresponds to the **time sequence**
 - What additional assumptions did you make?



- The observation depends directly on the state.
- Additional assumption
 - The next state depends only on the previous state (first order Markov assumption).

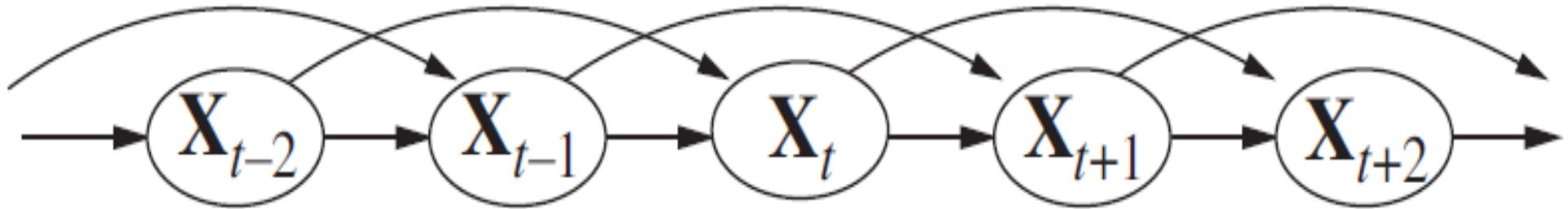
Markov Model

- The (first order) Markov assumption
- The distribution $P(X_t)$ depends only on the distribution $P(X_{t-1})$
 - $p(X_t|X_{t-1}, X_{t-2}, X_{t-3}, \dots) = p(X_t|X_{t-1})$
- The present (current state) can be predicted using local knowledge of the past (state at the previous step)

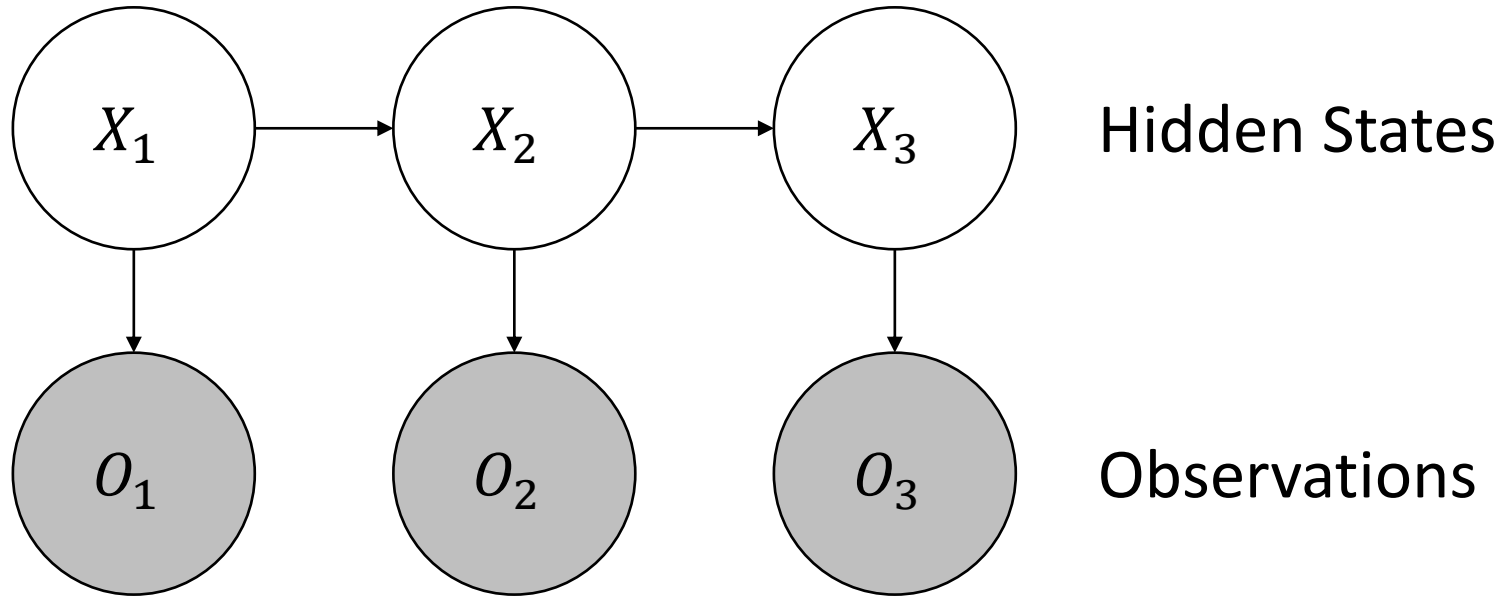


Second-order Markov Model

- State at time k depends on the states at times $k-1$ and $k-2$

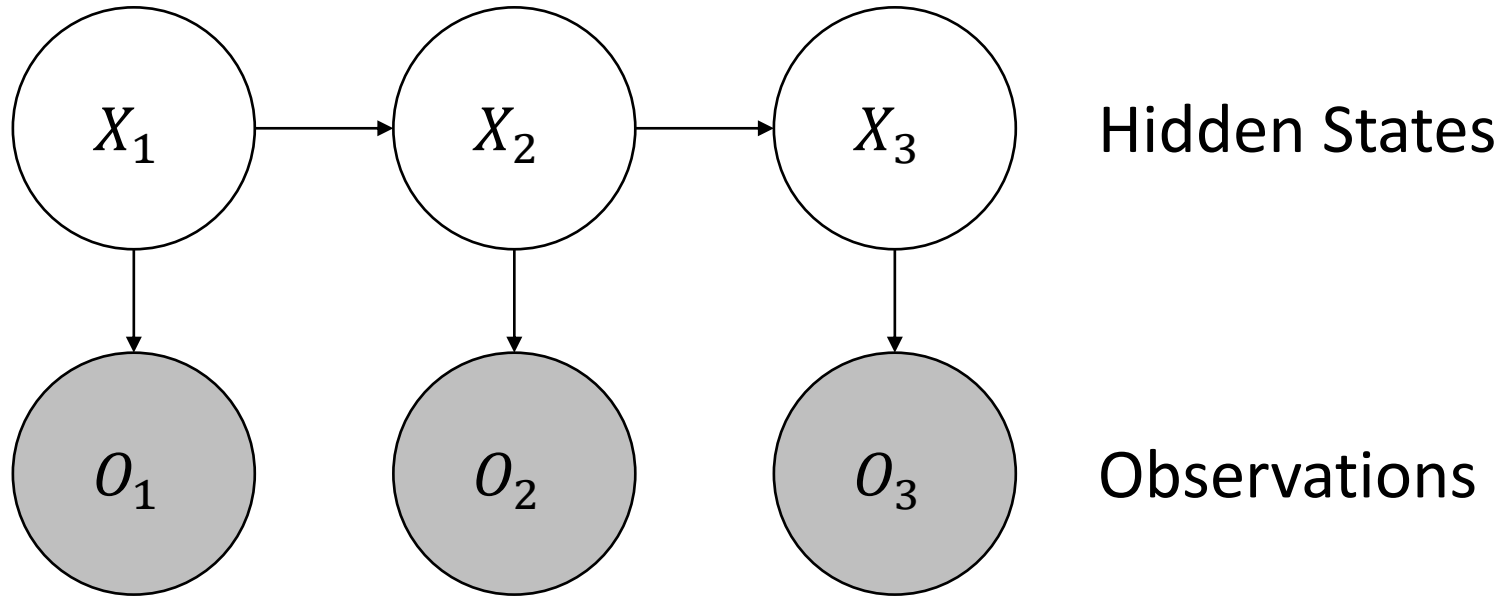


Hidden Markov Models (HMMs)



- Two important (conditional) independence properties:
 - Markov hidden process: future depends on the past via the present
 - Current observation independent of all else given current state

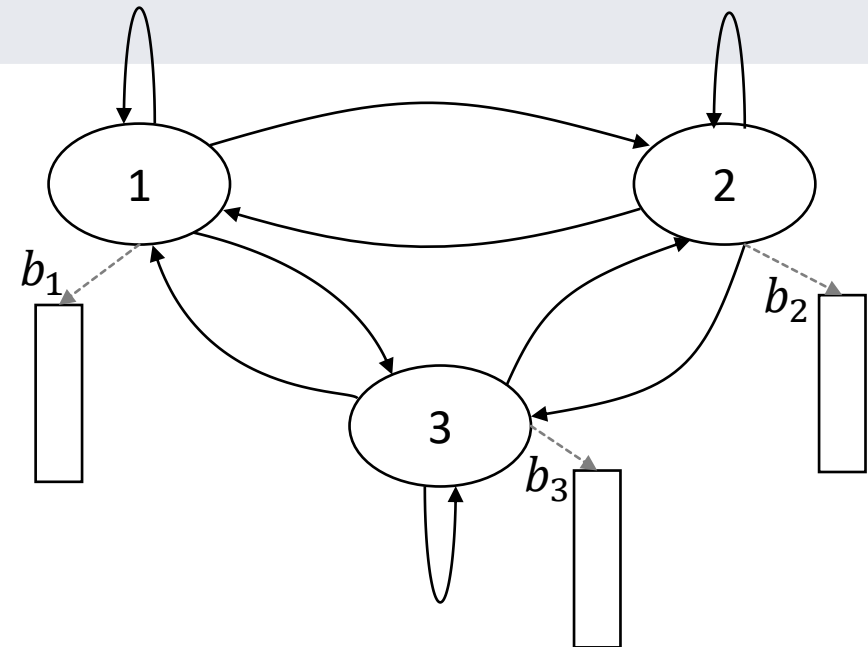
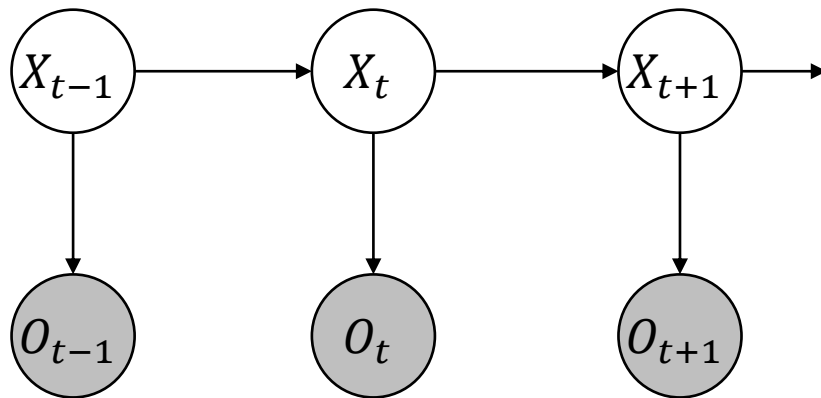
Hidden Markov Models (HMMs)



- State transition model: $P(X_t = j | X_{t-1} = i) = A(i, j) = a_{ij}$
- Observation model: $P(O_t = j | X_t = i) = b_{ij}$

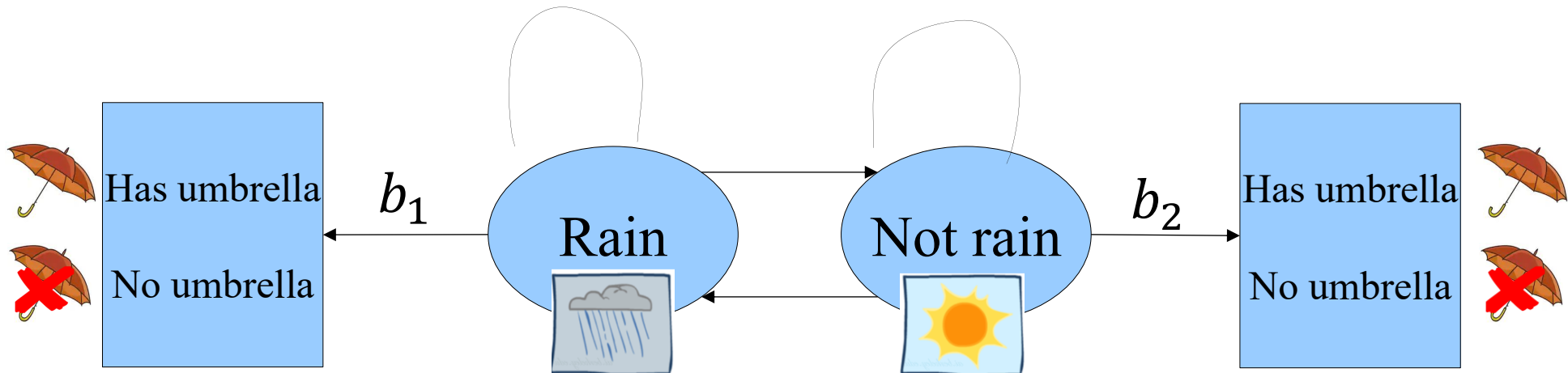
BN vs State Machine Representation

	Bayesian Network Notation	State Machine Representation
Arrows	Dependency between variables	State transitions and observation probs
Circles	Variables (states at each time step)	The different states



Example: Sequential Umbrella world

- A person tries to infer $rain = \{true, false\}$ **every day** by observing if a certain person has an *umbrella* = $\{true, false\}$ over consecutive days.
- Formulate using a State Machine Representation
 - How many states? Which?
Rain = true, Rain = False
 - How many observations? Which?
Umbrella = true, Umbrella = false



Elements of HMM

1. Number of states $N, x \in \{1, \dots, N\}$
2. Number of events $K, k \in \{1, \dots, K\}$
3. Initial State Probabilities
 $\pi = \{\pi_i\} = \{P(x_1 = i)\}$ for $1 \leq i \leq N$
4. State-transition probabilities
 $A = \{a_{ij}\} = \{P(x_t = j | x_{t-1} = i)\}$ for $1 \leq i, j \leq N$
5. Discrete Output Probabilities
 $B = \{b_i(k)\} = \{P(o_t = k | x_t = i)\}$ for $1 \leq i \leq N$ and $1 \leq k \leq K$

Elements of HMMs, π

- Initial Distribution
 - Contains the probability of the (hidden) model being in a particular hidden state at time $t = 1$ (sometimes $t = 0$).

- Often referred to as π

- Example:

$$\pi = [0.5 \ 0.2 \ 0.3], \text{ i.e.,}$$

$$p(X_1 = 1) = 0.5$$

$$p(X_1 = 2) = 0.2$$

$$p(X_1 = 3) = 0.3$$

Elements of HMMs, A

- State transition matrix A
 - holding the probability of transitioning from one hidden state to another hidden state.
- a_{21} is represented by $p(X_{t+1} = 1 | X_t = 2)$,
 - i.e., probability to transition from state 2 to state 1

$$\begin{array}{c|cccc} & X_{t+1} = 1 & X_{t+1} = 2 & \dots & X_{t+1} = N \\ \hline X_t = 1 & a_{11} & a_{12} & \dots & a_{1N} \\ X_t = 2 & \mathbf{a_{21}} & a_{22} & \dots & a_{2N} \\ \dots & \dots & \dots & \dots & \dots \\ X_t = N & a_{N1} & a_{N2} & \dots & a_{NN} \end{array} \quad \Bigg]$$

Elements of HMMs, B

- Output matrix B
 - probability of observing a particular measurement given that the hidden model is in a particular hidden state.
- $b_i(O_t = j)$ is represented by $p(O_t = j \mid X_t = i)$
 - i.e., probability to observe j in state i

$$\begin{bmatrix} & O_t = 1 & O_t = 2 & \dots & O_t = K \\ X_t = 1 & b_1(1) & b_1(2) & \dots & b_1(K) \\ X_t = 2 & b_2(1) & b_2(2) & \dots & b_2(K) \\ \dots & \dots & \dots & \dots & \dots \\ X_t = N & b_N(1) & b_N(2) & \dots & b_N(K) \end{bmatrix}$$

The model - λ

- λ sometimes just called M, is the model, i.e.,
- $\lambda = (A, B, \pi)$
 - A : state transition matrix
 - B : observation matrix, output matrix, emission probabilities, emissions
 - π : initial state distribution
- A , B and π are row-stochastic matrices (their rows sum to 1)

Prediction in HMMs

- Assume we have current belief $P(X \mid \text{evidence to date})$

$$P(X_t \mid O_{1:t})$$

- Then, after one time step passes:

$$p(X_{t+1} \mid O_{1:t}) = \{\text{sum rule}\} = \sum_{X_t} p(X_{t+1}, X_t \mid O_{1:t})$$

$$p(X) = \sum_Y P(X, Y)$$

$$= \{\text{product rule}\} = \sum_{X_t} p(X_{t+1} \mid X_t, O_{1:t}) p(X_t \mid O_{1:t})$$

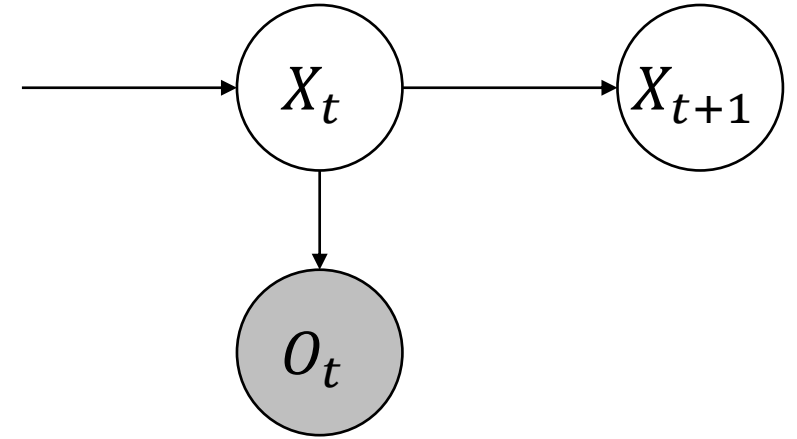
$$P(A, B) = P(A \mid B)P(B)$$

$$P(A, B \mid C) = P(A \mid B, C)P(B \mid C)$$

$$= \{O_{1:t} \text{ cond. indep of } X_{t+1} \text{ given } X_t\} = \sum_{X_t} p(X_{t+1} \mid X_t) p(X_t \mid O_{1:t})$$

$$X_{t+1} \perp O_{1:t} \mid X_t$$

- Basic idea: beliefs get “pushed” through the transitions



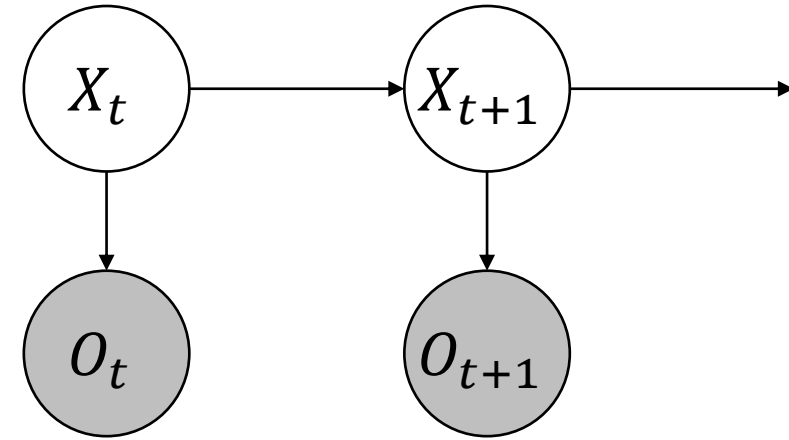
Update

- Assume we have current belief $P(X \mid \text{previous evidence})$

$$p(X_{t+1} \mid O_{1:t})$$

- Then, after evidence comes in:

$$p(X_{t+1} \mid O_{1:t+1}) = \{\text{"split" } O\} = p(X_{t+1} \mid O_{t+1}, O_{1:t})$$



$$= \{\text{Bayes rule, conditioned denominator}\} = \frac{p(O_{t+1} \mid X_{t+1}, O_{1:t}) p(X_{t+1} \mid O_{1:t})}{\sum_{X_{t+1}} p(O_{t+1} \mid X_{t+1}, O_{1:t}) p(X_{t+1} \mid O_{1:t})}$$

$$P(B|A) = \frac{P(A|B)P(B)}{\sum_B P(A|B)P(B)}$$

$$P(B|A, C) = \frac{P(A|B, C)P(B|C)}{\sum_B P(A|B, C)P(B|C)}$$

$$= \{O_{t+1} \text{ cond. indep of } O_{1:t} \text{ given } X_{t+1}\} = \frac{p(O_{t+1} \mid X_{t+1}) p(X_{t+1} \mid O_{1:t})}{\sum_{X_{t+1}} p(O_{t+1} \mid X_{t+1}) p(X_{t+1} \mid O_{1:t})}$$

$$O_{t+1} \perp O_{1:t} \mid X_{t+1}$$

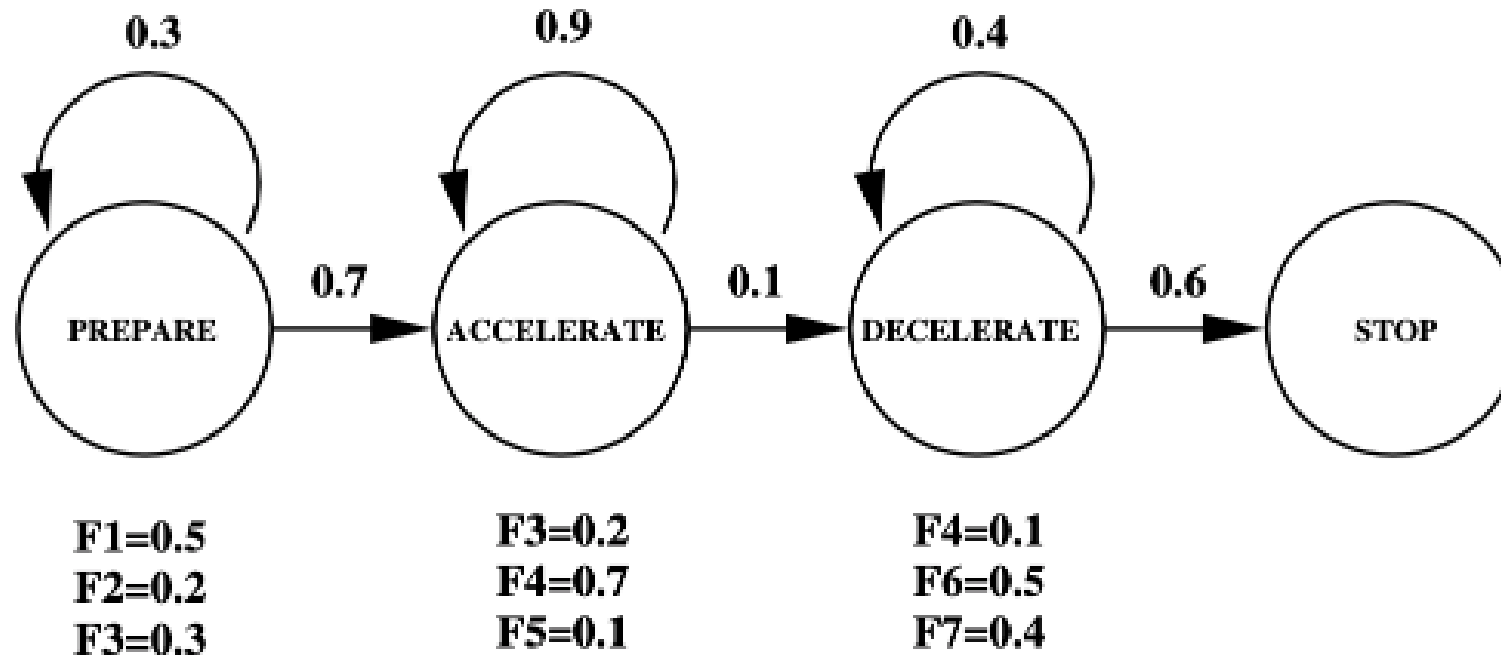
Example: Usain Bolt

- You have developed an automatic video annotation system for annotating recorded running sequences of Usain Bolt. The system takes images from a video stream of a running sequence as an input, it extracts some visual data and annotates each image as:
 - Usain is preparing for running
 - Usain runs/accelerates
 - Usain decelerates



Example: Usain Bolt

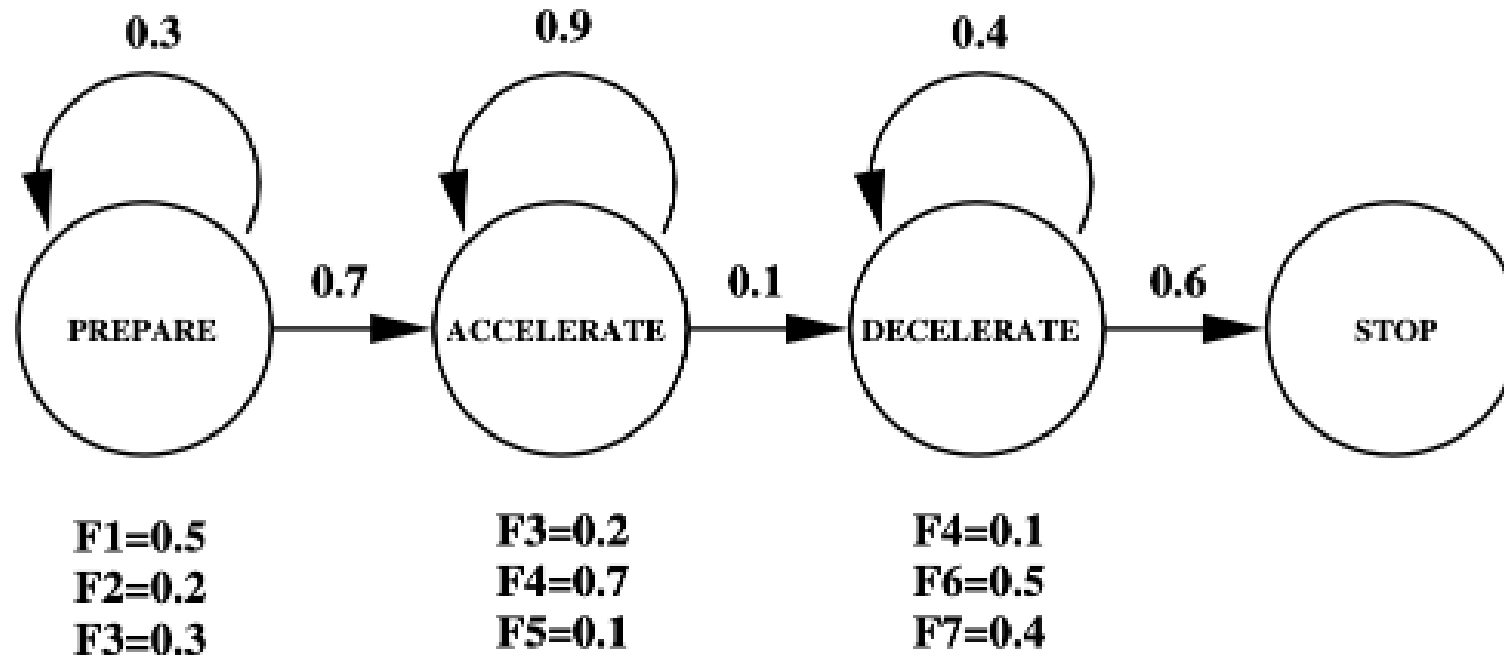
- F1 to 7 represent 7 ways of classifying frames
 - 7 possible observation outcomes that depict Usain's posture and probabilities of observing in each state
 - Observations not mentioned under a state have probability 0 of being observed there.
- STOP is a non-emitting terminal state, an extension of the standard HMM.
- Video sequences are divided in several video frames classified as F1 to F7.



Example: Usain Bolt

- What does A , B , and π look like?

$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0.5 & 0.2 & 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.7 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0.1 & 0 & 0.5 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \pi = [1 \quad 0 \quad 0 \quad 0]$$



Example: Usain Bolt

- Given:

$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0.5 & 0.2 & 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.7 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0.1 & 0 & 0.5 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \pi = [1 \quad 0 \quad 0 \quad 0]$$

- How to calculate $P(X_2 = ACCELERATE)$?
 - Start with π and propagate probability one step with A
 - $\pi = X_1 = [1 \ 0 \ 0 \ 0] \rightarrow X_2 = \pi A = [0.3 \ 0.7 \ 0 \ 0]$
 - $p(X_2 = ACCELERATE) = 0.7$

Example: Usain Bolt

- Given:

$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0.5 & \boxed{0.2} & 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.7 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0.1 & 0 & 0.5 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \pi = [1 \quad 0 \quad 0 \quad 0]$$

- How to calculate $P(O_2 = F_2)$?
 - We know $P(X_2) = [0.3 \ 0.7 \ 0 \ 0]$
 - And we know how probable each measurement is in each state from B. Use sum rule.

$$\begin{aligned} p(O_2 = F_2) &= \sum_{i=1}^N p(O_2 = F_2 \mid X_2 = i) p(X_2 = i) \\ &= 0.2 \cdot 0.3 + 0 \cdot 0.7 + 0 \cdot 0 + 0 \cdot 0 = 0.06 \end{aligned}$$

Example: Usain Bolt

- Given:

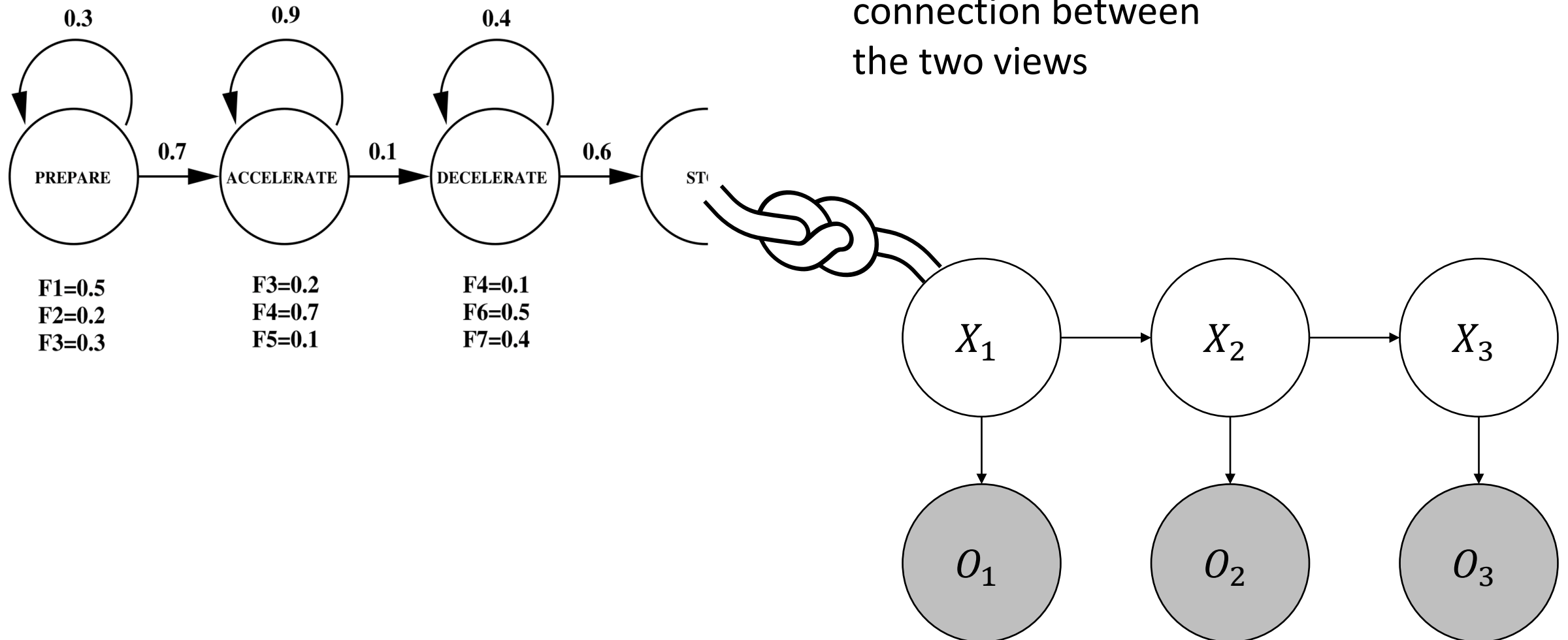
$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad B = \begin{matrix} & F_2 \\ \begin{bmatrix} 0.5 & 0.2 & 0.3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.7 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0.1 & 0 & 0.5 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad \pi = [1 \quad 0 \quad 0 \quad 0]$$

- How to calculate $P(X_2 = ACCELERATE | O_2 = F_2)$?
 - From previous slides $p(O_2 = F_2) = 0.06$ and $p(X_2 = ACCELERATE) = 0.7$
 - Use Bayes rule

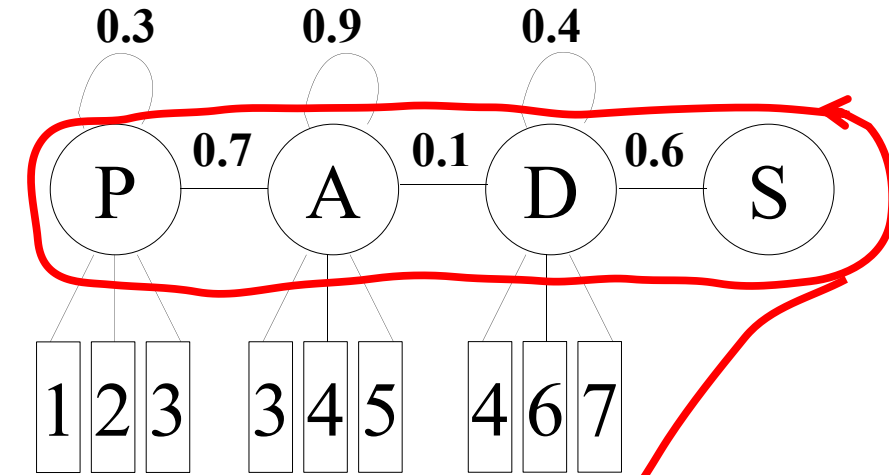
$$p(X_2 = ACCELERATE | O_2 = F_2) = \frac{p(O_2 = F_2 | X_2 = ACCELERATE)p(X_2 = ACCELERATE)}{p(O_2 = F_2)} = \frac{0 \cdot 0.7}{0.06} = 0$$

- Could we have seen this immediately?
 - YES. Cannot measure F2 in Accelerate state, i.e., $p=0$

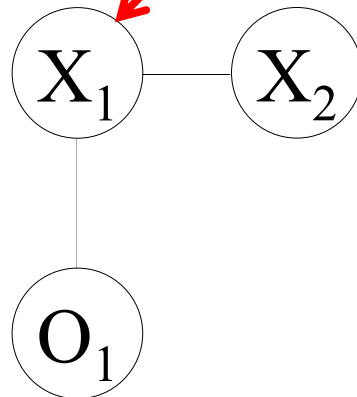
Example: Usain Bolt



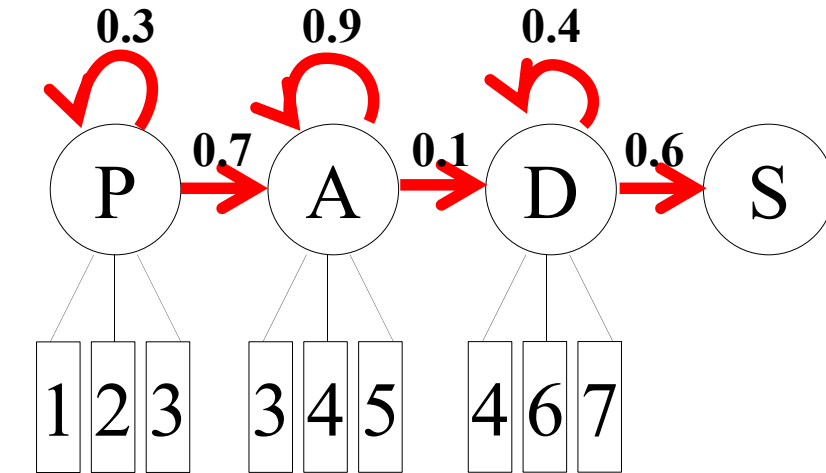
The states



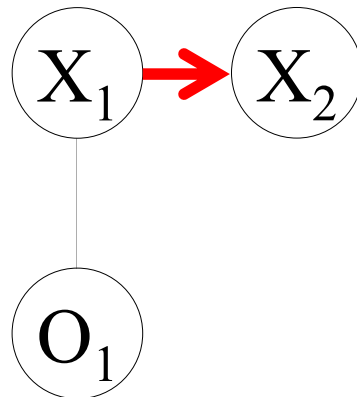
The states that variables $X_1, X_2, \dots$ can be in (remember discrete)



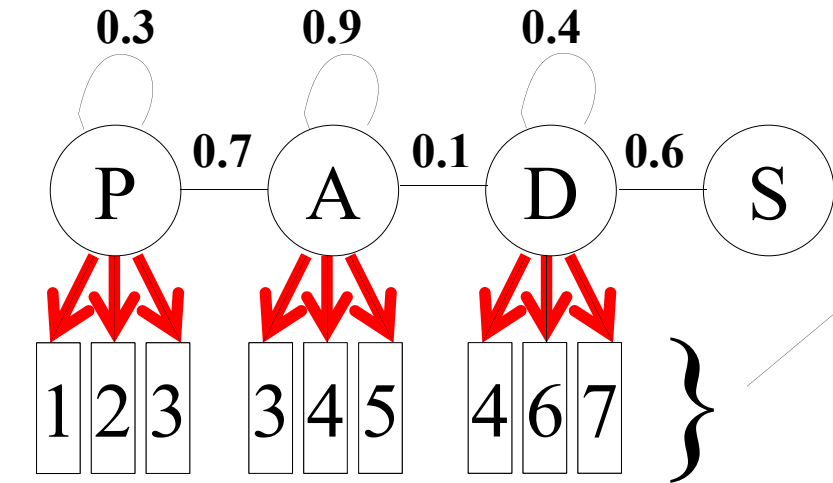
The state transitions



Red lines describe $p(x_{k+1} | x_k)$
i.e., the state transitions.
Captured with the A matrix (top)
and $p(x_{k+1} | x_k)$ (bottom)

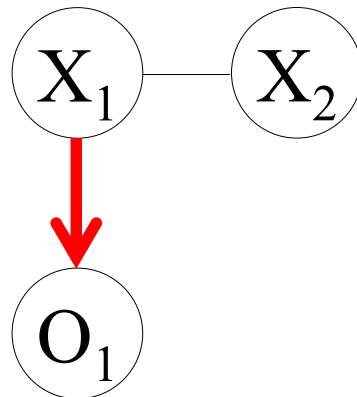


The observation/emission model

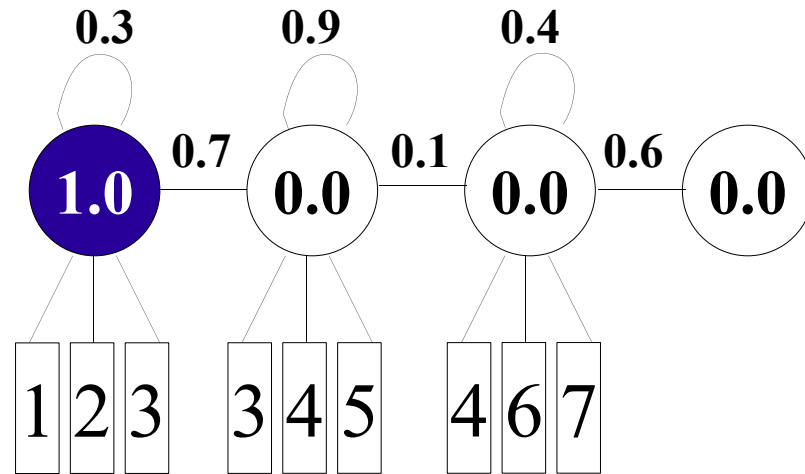


Different possible observations,
(aka “symbols”)

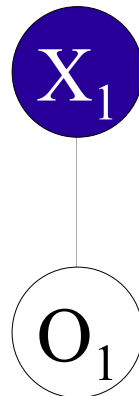
Red lines describe $p(O_k | X_k)$
i.e., observation model.
Captured by matrix B (top)
and $p(O_k | X_k)$ (bottom).



Initial state $\pi = [1 \ 0 \ 0 \ 0]$



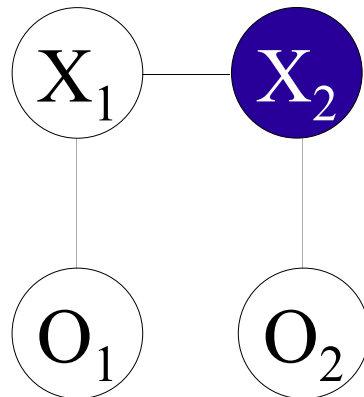
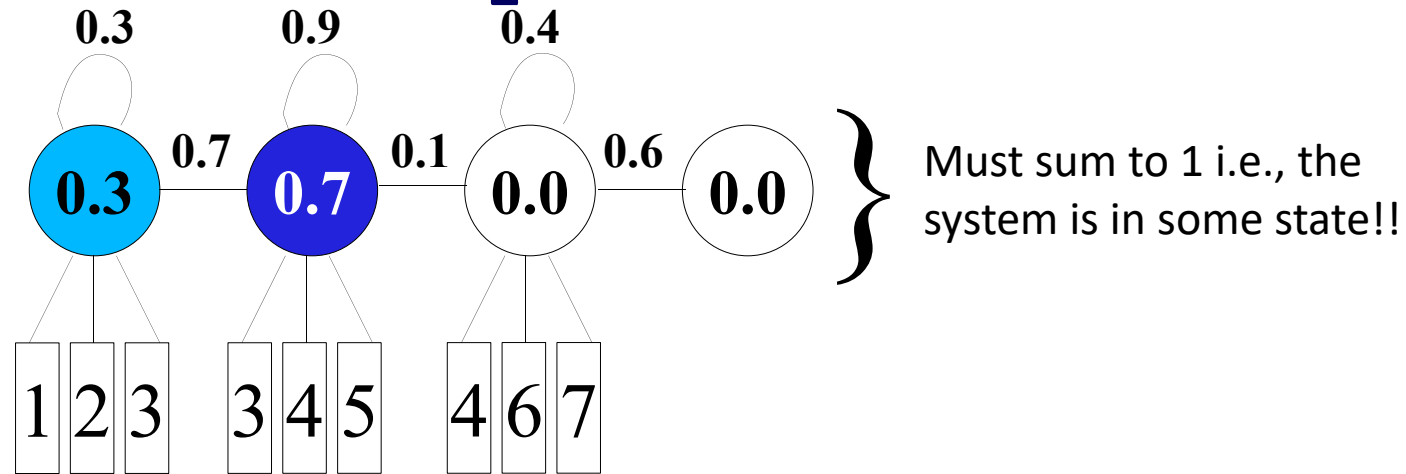
The darker the higher probability



$$\pi = [1 \quad 0 \quad 0 \quad 0]$$

$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Prediction for $X_2 (= \pi A)$

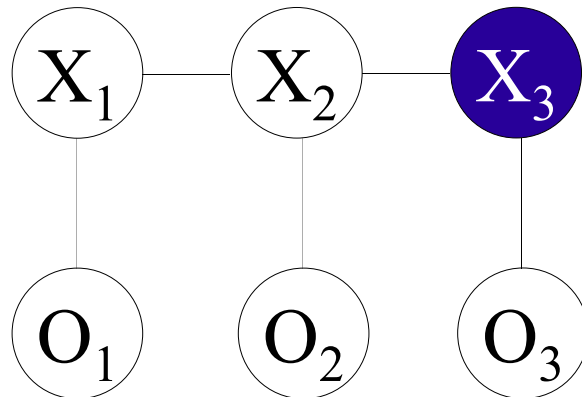
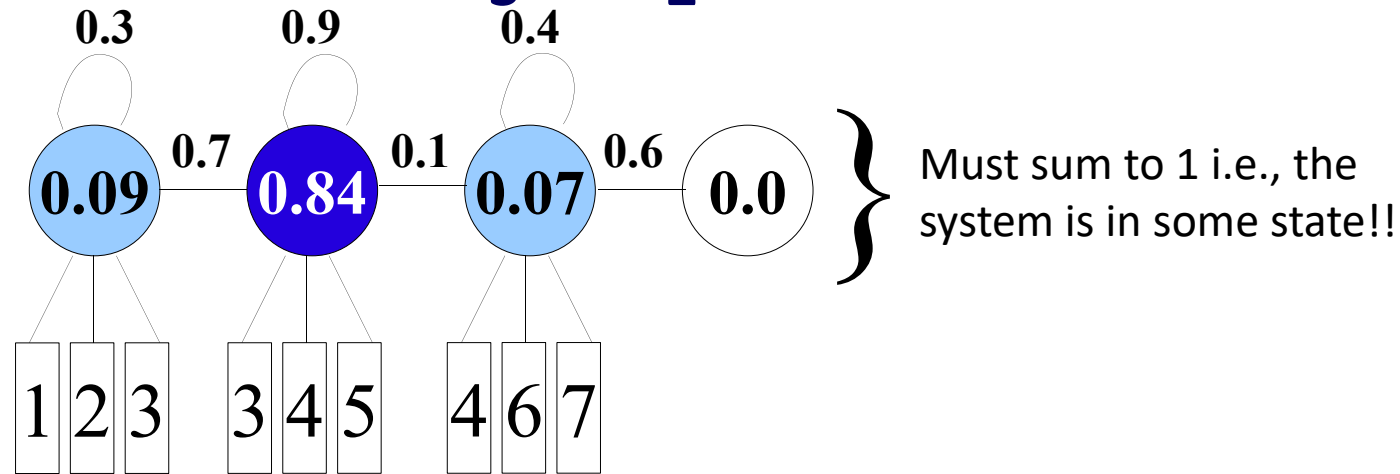


$$\pi = [1 \quad 0 \quad 0 \quad 0]$$

$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\pi A = X_2 = [0.3 \quad 0.7 \quad 0 \quad 0]$$

Prediction for $X_3 (=X_2A)$

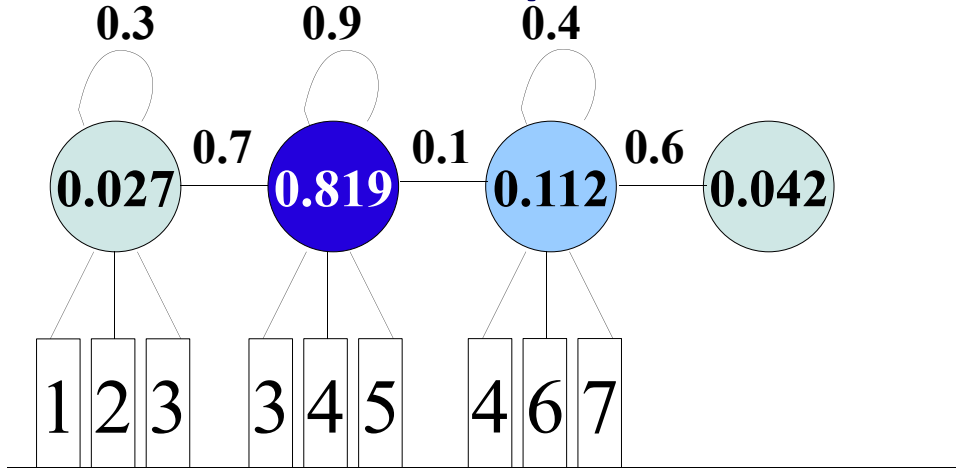


$$X_2 = [0.3 \quad 0.7 \quad 0 \quad 0]$$

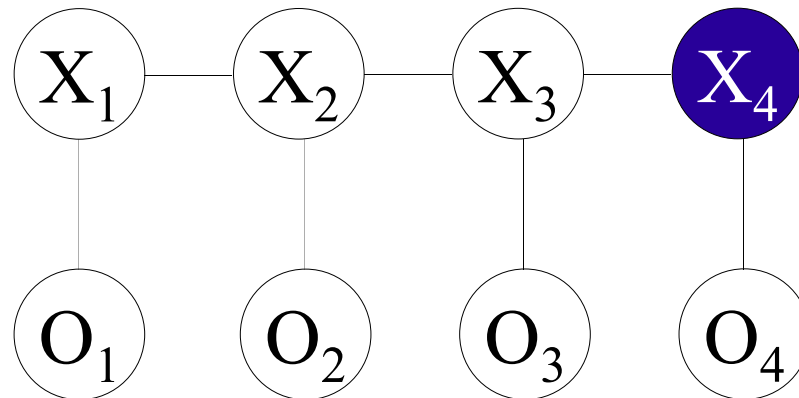
$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$X_2 A = X_3 = [0.09 \quad 0.84 \quad 0.07 \quad 0]$$

Prediction for X_4



Note: We know for sure that we are at $t=4$ but not in which state we are. This is what we often try to estimate!



$$X_3 = [0.09 \quad 0.84 \quad 0.07 \quad 0]$$

$$A = \begin{bmatrix} 0.3 & 0.7 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

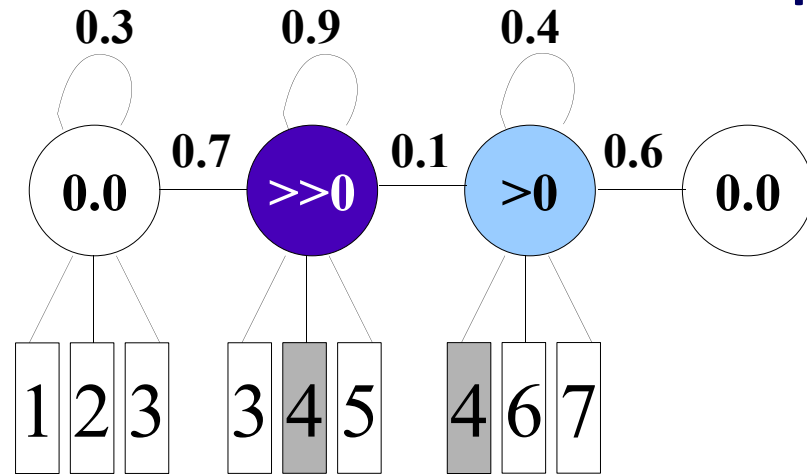
$$X_3 A = X_4 = [0.027 \quad 0.819 \quad 0.112 \quad 0.042]$$

Measurements?

- Without measurements we are doing pure prediction
 - can do that ahead of time!
- Measurements give clues to what state we are in
 - beliefs get reweighted, observations can reduce uncertainty



Assume we measure $O_4 = 4$ (i.e. obs 4 at $t=4$)



- We have $p(X_4)$ from the prediction
- For each state j we can calculate $p(X_4 = j | O_4)$

$$p(X_4 = j | O_4) = \{Bayesrule\}$$

$$= \frac{p(O_4 = 4 | X_4 = j)p(X_4 = j)}{p(O_4)}$$

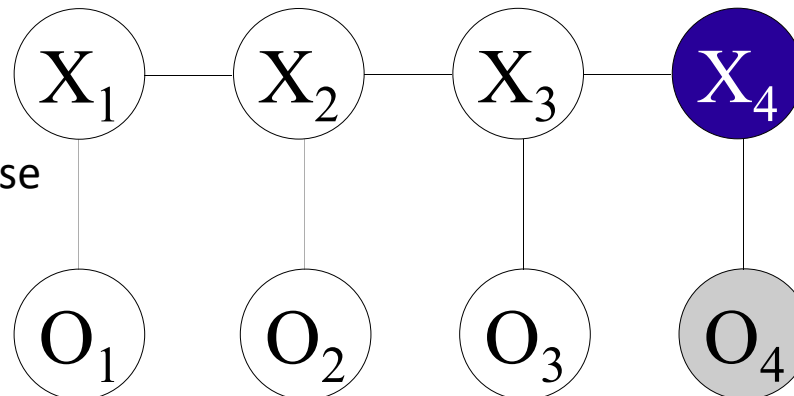
$$= \frac{b_j(4)p(X_4 = j)}{p(O_4)}$$

$$= \eta b_j(4)p(X_4 = j)$$

$$with \eta = \frac{1}{\sum_j b_j(4) \times p(X_4 = j)}$$

Weight with $p(O|X)$

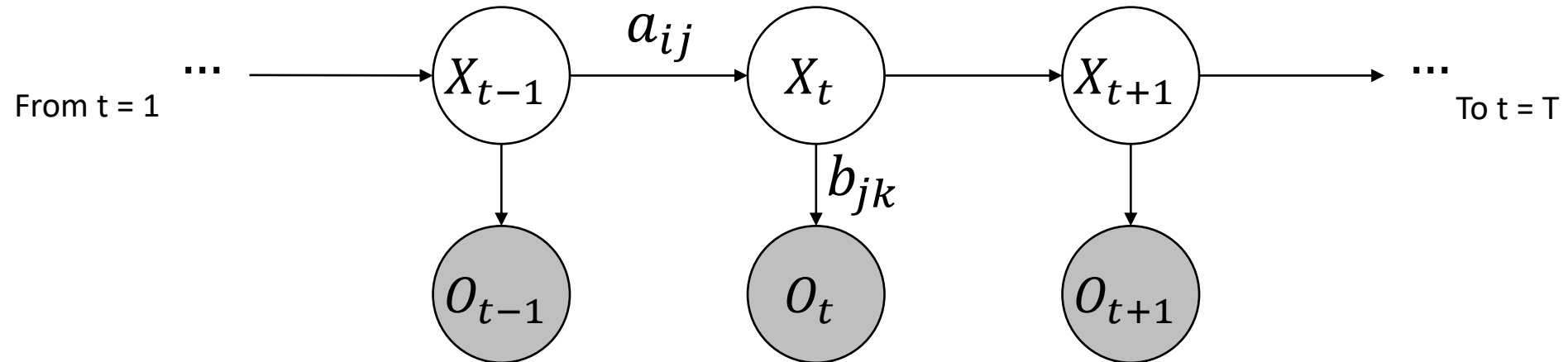
→ not in state 1 for sure because we cannot measure 4 there



Observed (grey)

HMM Terminology

Time instants	$t \in \{1, 2 \dots T\}$
Hidden States / States / Emitters	X_t
Outputs / Emissions / Observations / Visible States	O_t
All possible states / states set	$X_t \in \{1, 2, \dots, N\}$
All possible emissions / emissions set	$O_t \in \{1, 2, \dots, K\}$
Initial state distribution / Initial state probabilities	p_i in q or π_i in π
Transition probabilities / State transition probabilities	a_{ij} in row – stochastic matrix A
Emission probabilities / Observation probabilities	b_{jk} in row – stochastic matrix B



Three problems solved with HMMs

- 1. Evaluation:** Compute likelihood $P(O_{1:t}|\lambda)$
of observation sequence $(O_{1:T}=\{O_1, O_2, \dots, O_t, \dots, O_T\})$ given λ
Forward algorithm
Backward algorithm
- 2. Decoding:** Most likely state sequence $X^*_{1:T}$ given $O_{1:T}$ and λ
Viterbi algorithm
- 3. Learning:** Estimate model parameters
 $\Lambda=\{\lambda\}$ given $O_{1:T}$ such that $p(O_{1:T}|\lambda)$ is maximized
 $\rightarrow$ A and B matrices and π
Baum-Welch algorithm

1. Compute likelihood $p(\mathbf{O}_{1:t} | \lambda)$ of observed sequence given λ

- Given:

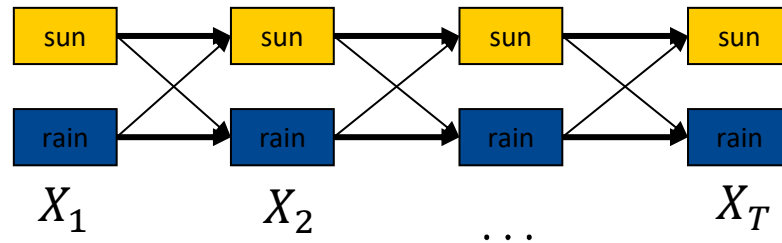
- A, B, π

- Emission sequence $\mathbf{O} = \{O_1, O_2 \dots O_T\}$



- Unknown:

- Hidden state sequence $\mathbf{X} = \{X_1, X_2 \dots X_T\}$ that actually produced $\mathbf{O}$.



- To Find:

- Probability that the given sequence $\mathbf{O}$ occurred **regardless of which $\mathbf{X}$ produced the sequence.**

Likelihood of $p(O_{1:T} | \lambda)$

$$\begin{aligned} p(O_{1:T} | \lambda) &= \{\text{sum rule}\} = \sum_{X_{1:T}} p(O_{1:T}, X_{1:T}) \\ &= \{\text{product rule}\} = \sum_{X_{1:T}} p(O_{1:T} | X_{1:T}) p(X_{1:T}) \\ &= \sum_{X_{1:T}} \pi_{X_1} \underbrace{a_{X_1 X_2} a_{X_2 X_3} \dots a_{X_{T-1} X_T}}_{\substack{p(X_{1:T}) \\ \text{Transition Probabilities}}} \underbrace{b_{X_1}(O_1) b_{X_2}(O_2) \dots b_{X_T}(O_T)}_{\substack{p(O_{1:T} | X_{1:T}) \\ \text{Emission Probabilities}}} \end{aligned}$$

- Note that we are summing over all N^T permutations of $X_{1:T}$
- Evaluating this requires $O(N^T)$ operations — exponential in T
- Can be formulated recursively using the forward (and backward) algorithm

Forward algorithm (aka α -pass)

- Introduce:

$$\alpha_t(i) = p(O_{1:t}, X_t = i \mid \lambda) \forall t = 1, \dots, T$$

- Initialize as:

$$\alpha_1(i) = \pi_i b_i(O_1)$$

$\alpha_t(i)$ is the probability of observing a partial sequence of observables $O_1, \dots, O_t$ AND at time t , being in state i

Forward algorithm (aka α -pass)

- Introduce:

$$\alpha_t(i) = p(O_{1:t}, X_t = i \mid \lambda) \forall t = 1, \dots, T$$

- Initialize as:

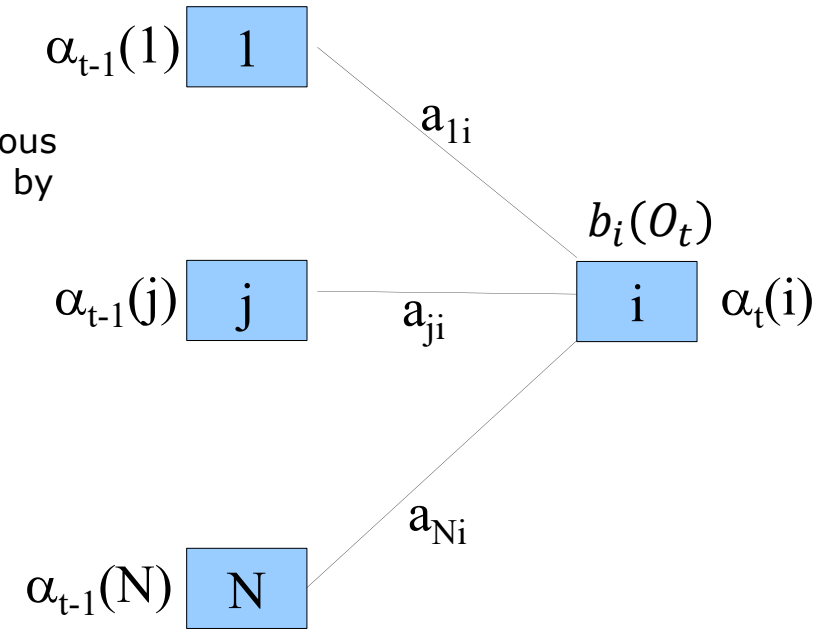
$$\alpha_1(i) = \pi_i b_i(O_1)$$

- For $2 \leq t \leq T$:

$$\alpha_t(i) = \left[\sum_{j=1}^N \alpha_{t-1}(j) a_{ji} \right] b_i(O_t)$$

Prediction (sum of the previous partial probabilities multiplied by the transition probabilities)

Weight (observation probability)



Forward algorithm (aka α -pass)

- Introduce:

$$\alpha_t(i) = p(O_{1:t}, X_t = i \mid \lambda) \forall t = 1, \dots, T$$

- Initialize as:

$$\alpha_1(i) = \pi_i b_i(O_1)$$

Prediction (sum of the previous partial probabilities multiplied by the transition probabilities)

Weight (observation probability)

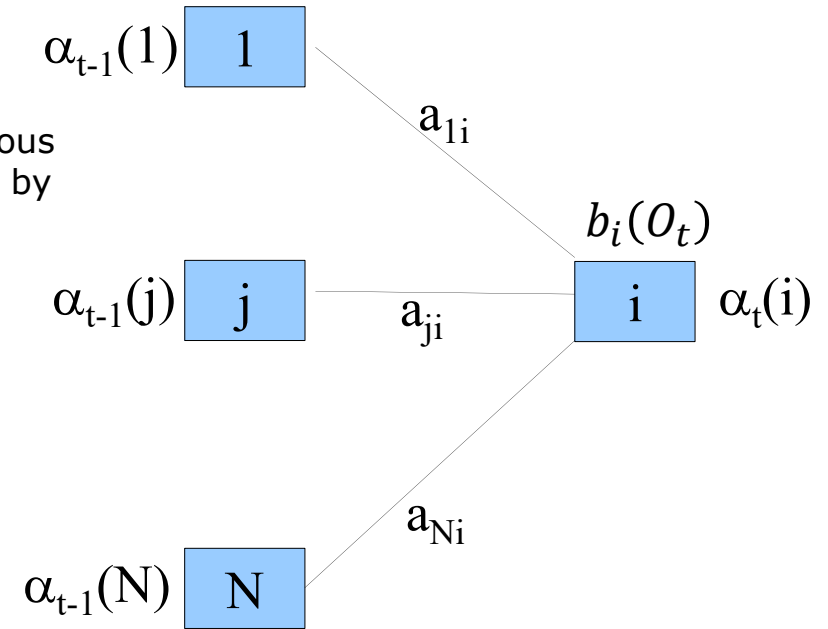
- For $2 \leq t \leq T$:

$$\alpha_t(i) = \left[\sum_{j=1}^N \alpha_{t-1}(j) a_{ji} \right] b_i(O_t)$$

- Which gives us:

$$p(O_{1:T} \mid \lambda) = \sum_{i=1}^N p(O_{1:T}, X_t = i \mid \lambda) = \sum_{i=1}^N \alpha_T(i)$$

- recursive way to calculate likelihood with only N^2T multiplications (compared to N^T)



Forward algorithm intuition

O_1



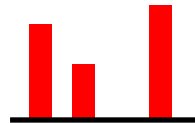
O_2



O_3



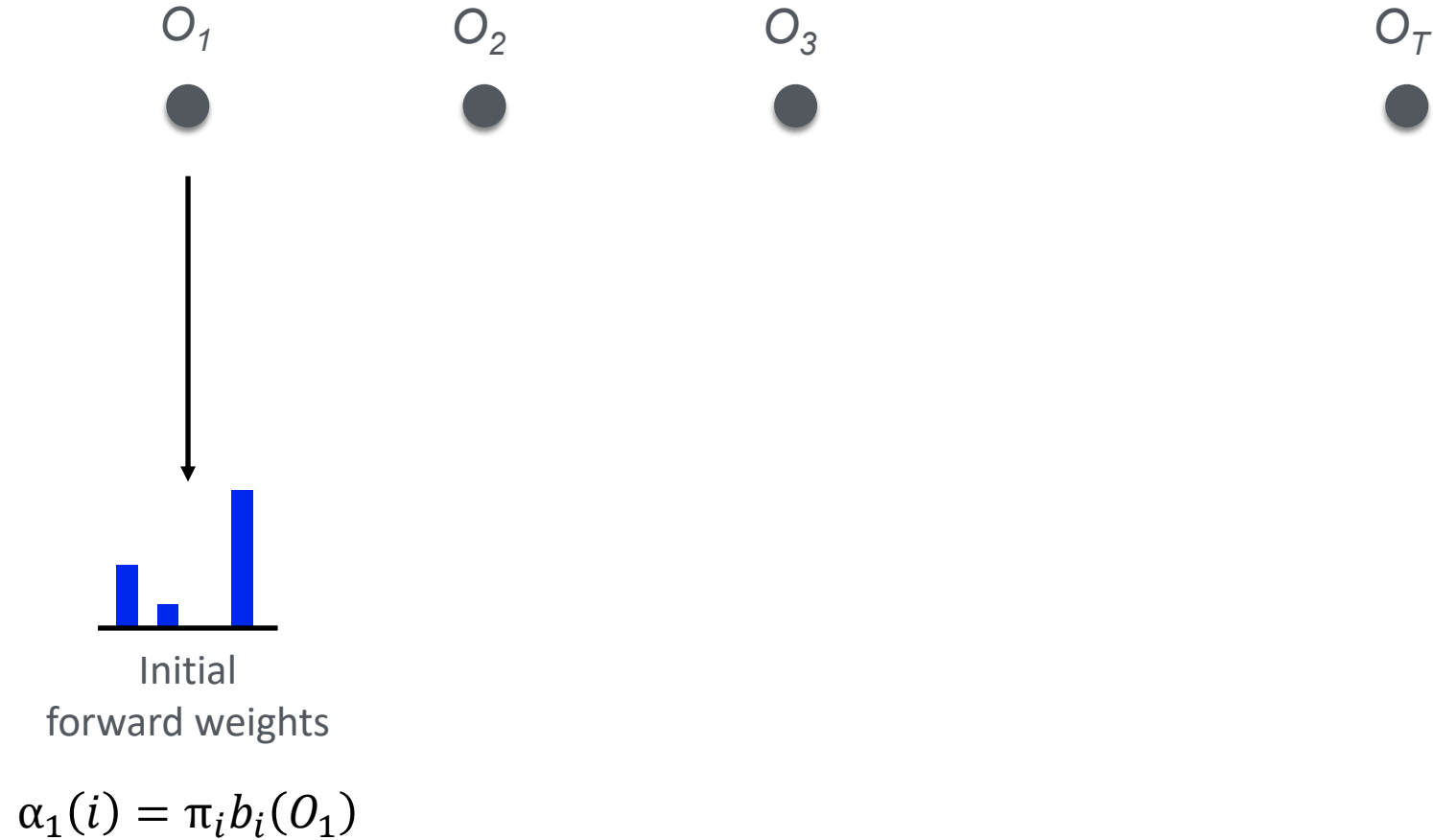
O_T



Initial
distribution

π_i

Forward algorithm intuition



Forward algorithm intuition

O_1



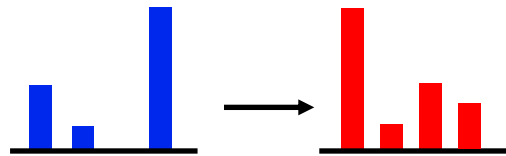
O_2



O_3



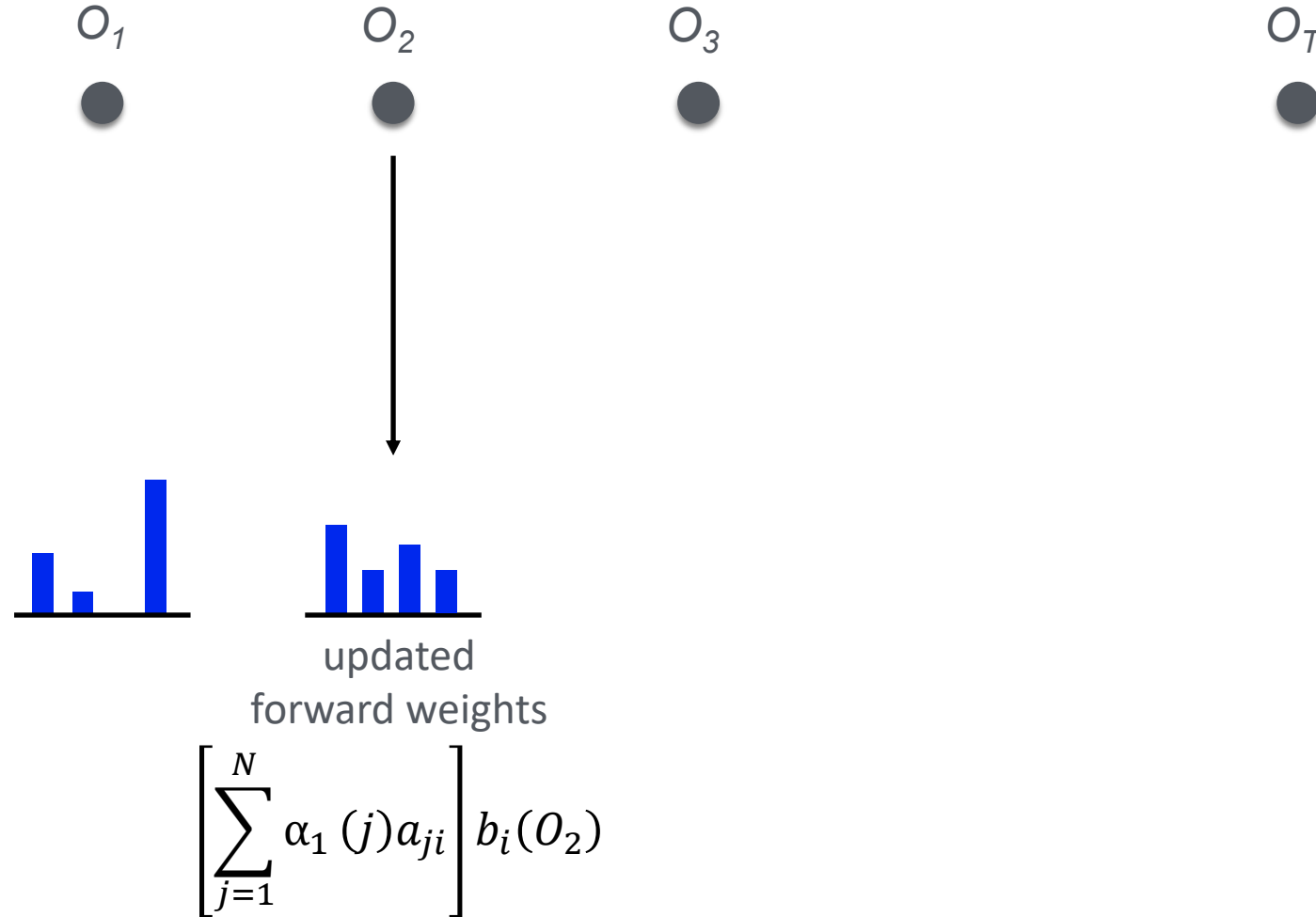
O_T



Predicted
weights

$$\left[\sum_{j=1}^N \alpha_1(j) a_{ji} \right]$$

Forward algorithm intuition



Forward algorithm intuition

O_1



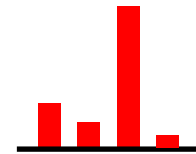
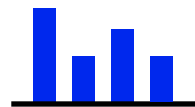
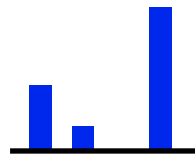
O_2



O_3



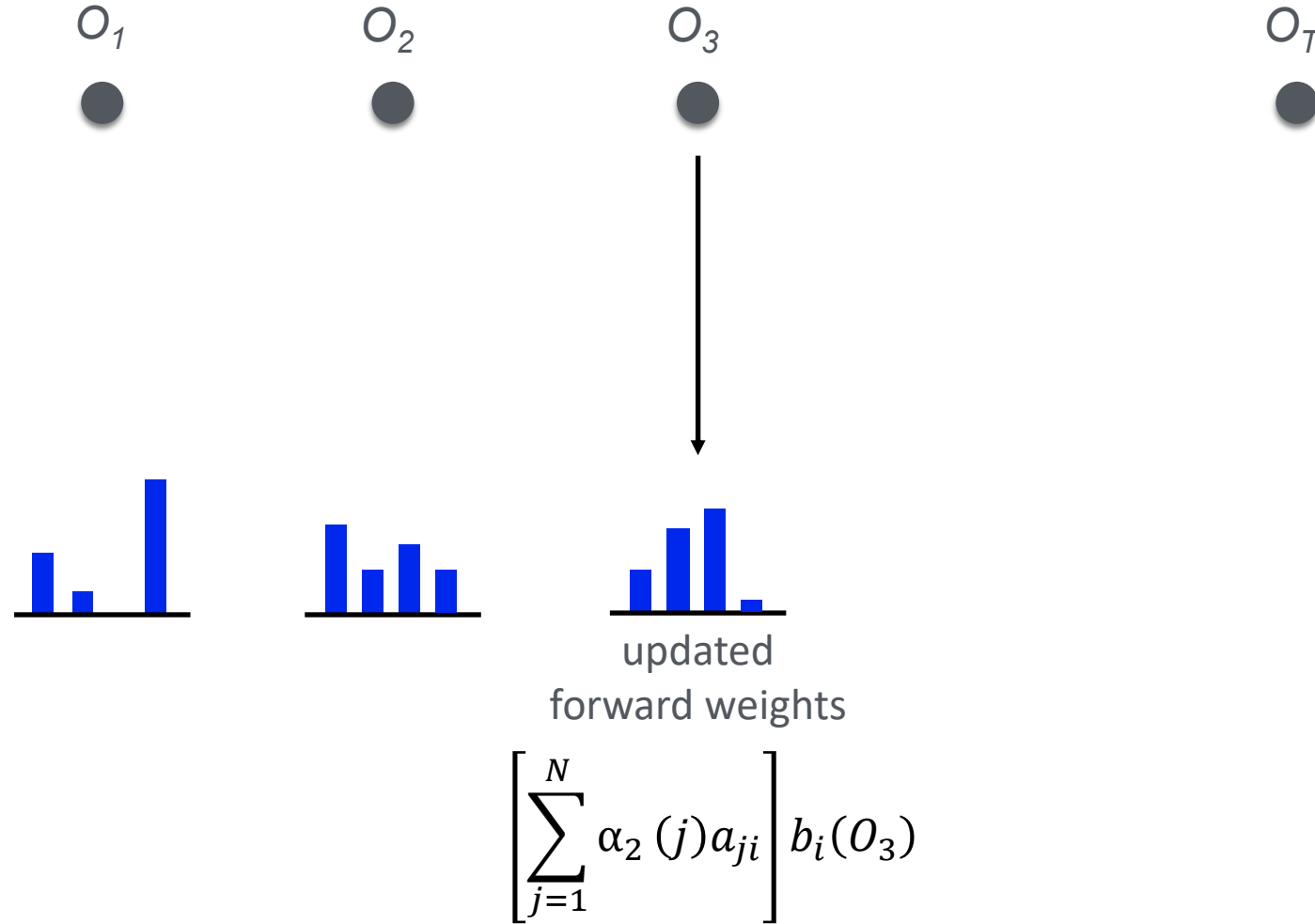
O_T



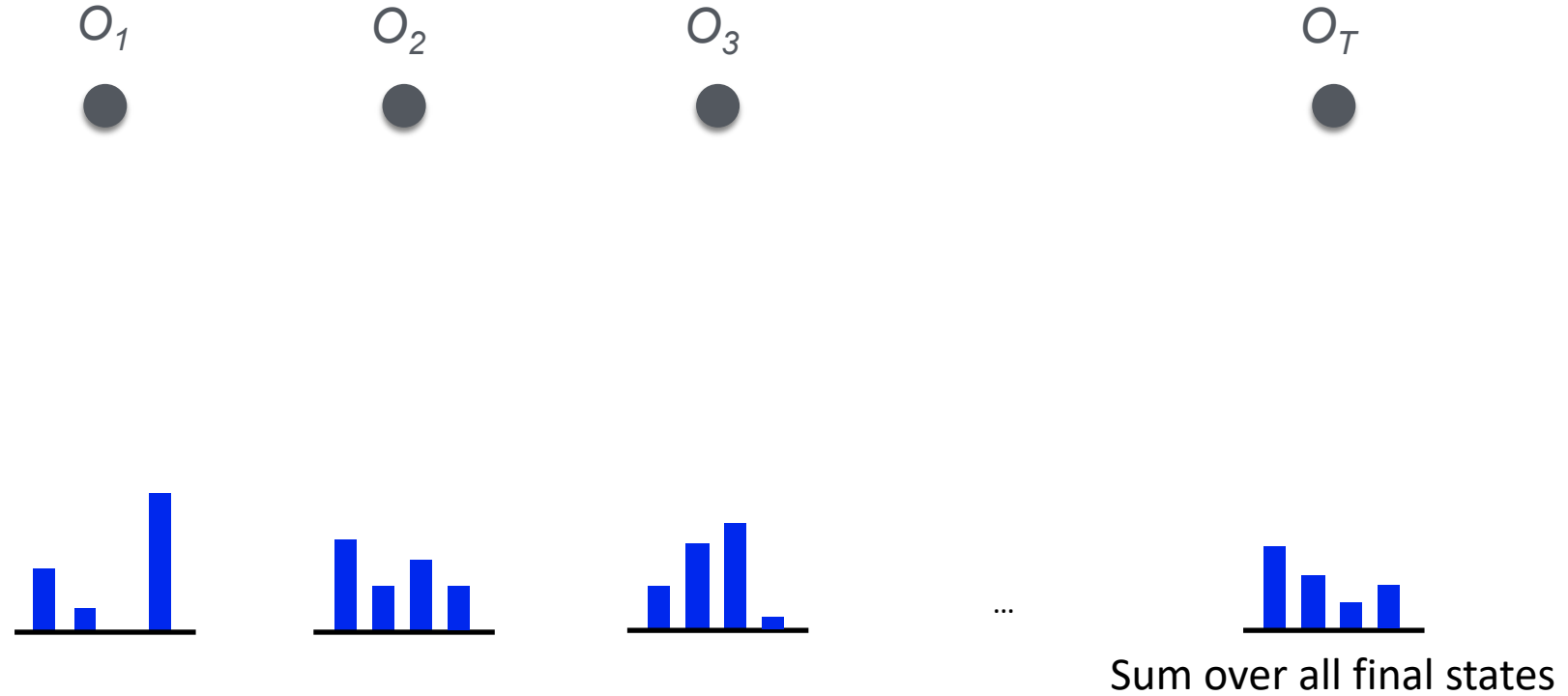
Predicted
weights

$$\left[\sum_{j=1}^N \alpha_2(j) a_{ji} \right]$$

Forward algorithm intuition



Forward algorithm intuition



$$P(O_{1:T} \mid \lambda) = \sum_{i=1}^N \alpha_T(i)$$

Backward algorithm (aka β -pass)

- $\beta_t(i)$: probability of the remaining observations $O_{t+1}, \dots, O_T$ given that the hidden state at time t is i .

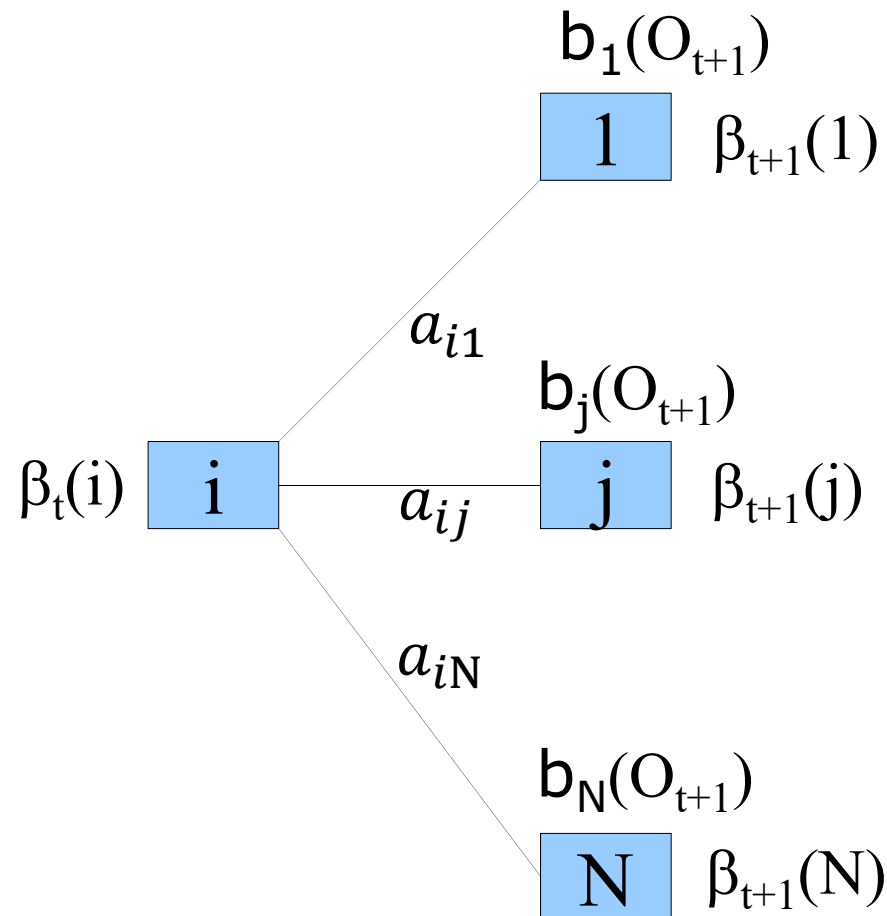
- **Introduce:**
$$\beta_t(i) = p(O_{t+1:T} \mid X_t = i, \lambda)$$

- **Initialize:**
$$\beta_T(i) = 1, \forall i = 1, \dots, N$$

- **For $t = T-1, T-2, \dots, 1$.**
$$\beta_t(i) = \sum_{j=1}^N a_{ij} b_j(O_{t+1}) \beta_{t+1}(j) \longrightarrow \text{How the present relates to the future}$$

One step in the backward algorithm

$$\beta_t(i) = \sum_{j=1}^N a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)$$



Arrows show the direction of computation; a_{ij} still describes the transition from i to j .

Three problems solved with HMMs

- 1. Evaluation:** Compute likelihood $P(O_{1:t}|\lambda)$
of observation sequence $(O_{1:T}=\{O_1, O_2, \dots, O_t, \dots, O_T\})$ given λ
Forward algorithm
Backward algorithm
- 2. Decoding:** Most likely state sequence $X^*_{1:T}$ given $O_{1:T}$ and λ
Viterbi algorithm
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Baum-Welch algorithm

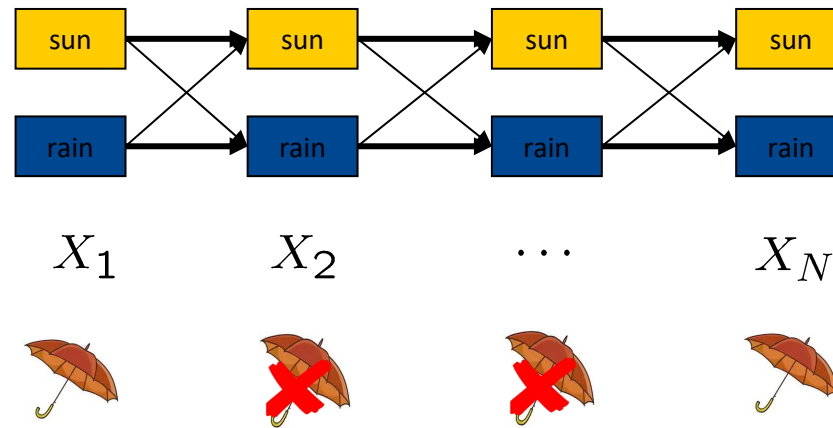
2. Calculate most likely state sequence

- Motivating examples
 - Parts-Of-Speech (POS) tagging:
 - Given a sentence such as “I love cats and dogs”
 - Find POS tags <pronoun><verb><noun><conjunction><noun>
 - Speech recognition
 - Given a sound recording of spoken words
 - Find which words were spoken
- “Recognize speech”
“Wreck a nice beach”
- HMM will return the **most likely** sequence of words (hidden states)



2. Calc most likely state sequence

- Given:
 - Emission sequence $\mathbf{O} = \{O_1, O_2 \dots O_T\}$
 - A, B, π
- To Find:
 - Hidden state sequence $\mathbf{X}^* = \{X_1, X_2 \dots X_T\}$ that most likely produced $\mathbf{O}$.
 - Probability of occurrence of $\mathbf{X}^*$



2. Calculate most likely state sequence

- Select the hidden-state sequence with the highest probability given the observations.

$$X_{1:T}^* = \underset{X_{1:T}}{argmax} p(X_{1:T} | O_{1:T}, \lambda)$$

- Cannot solve individually for each time step
- Need to optimize the sequence not just individual states

Forward vs. Viterbi algorithm

- Forward algorithm computes sums of paths, Viterbi computes best paths

$$\alpha_t(i) = \left[\sum_{j=1}^N \alpha_{t-1}(j) a_{ji} \right] b_i(O_t)$$

Forward algorithm (Sum)

$$\delta_t(i) = \max_{j \in \{1, \dots, N\}} [\delta_{t-1}(j) a_{ji}] b_i(O_t)$$

Viterbi algorithm (Max)

Most likely sequence (Viterbi alg.)

- Solved using dynamic programming (DP) with an algorithm called Viterbi.

- Initialize:

$$\delta_1(i) = \pi_i b_i(O_1), i = 1, \dots, N$$

- For each $t > 1$:

$$\delta_t(i) = \max_{j \in \{1, \dots, N\}} [\delta_{t-1}(j) a_{ji}] b_i(O_t)$$

Partial prob of one of the
most probable paths to
state i at time t

- Probability of best path:

$$\max_{j \in \{1, \dots, N\}} [\delta_T(j)]$$

- Find path by keeping book of preceding states and trace back from highest-scoring final state : $\Psi_{t(i)} = \underset{j \in \{1, \dots, N\}}{\operatorname{argmax}} [\delta_{t-1}(j) a_{ji}]$

2. Calc most likely state sequence

O_1



O_2



O_3



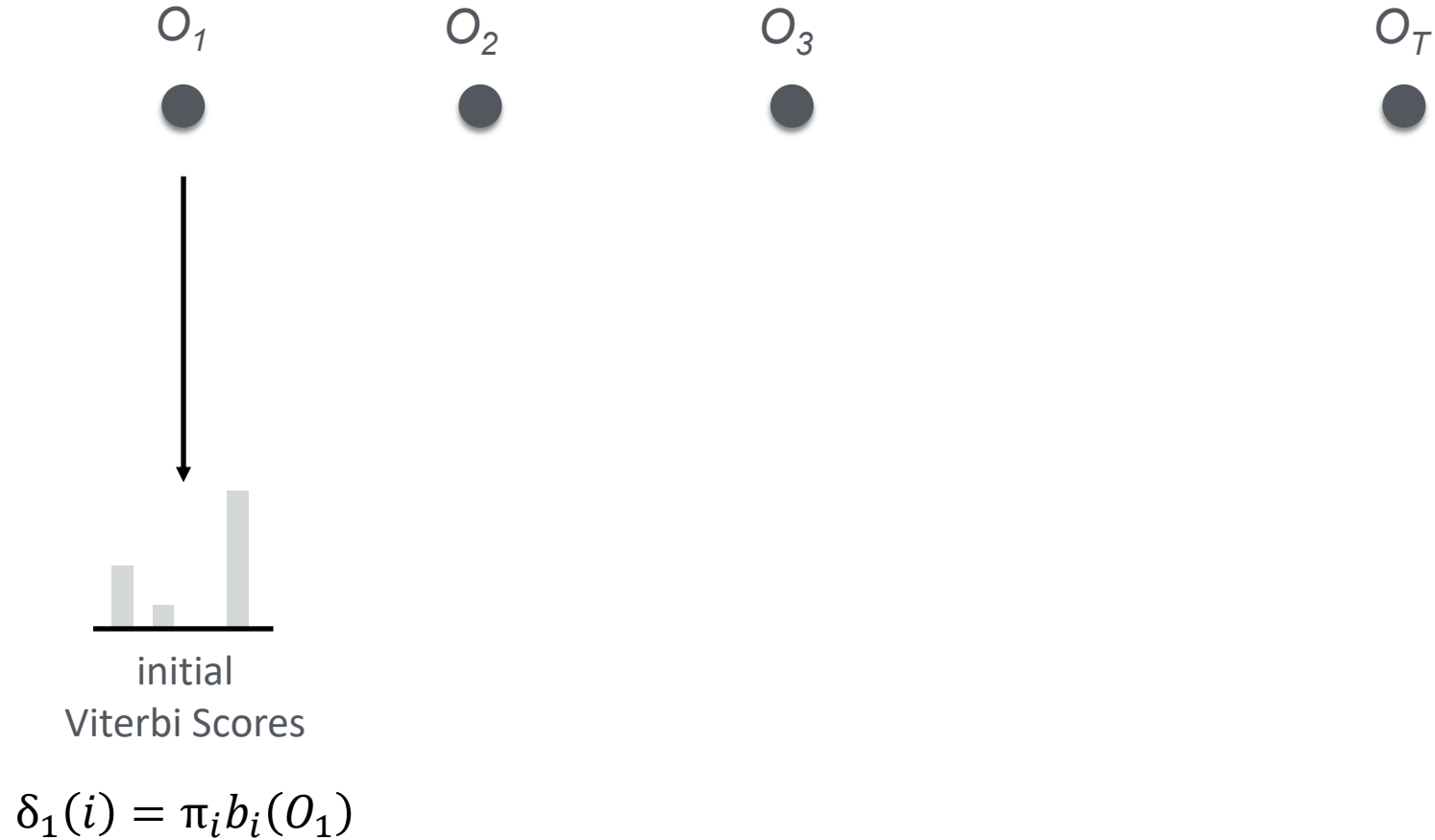
O_T



Initial
distribution

π_i

2. Calc most likely state sequence



2. Calc most likely state sequence

O_1



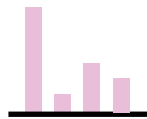
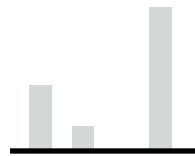
O_2



O_3



O_T



best incoming
scores.

$$\max_{j=1,\dots,N} [\delta_1(j) a_{ji}]$$

2. Calc most likely state sequence

O_1



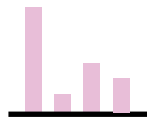
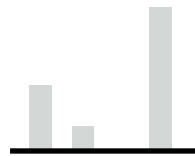
O_2



O_3



O_T

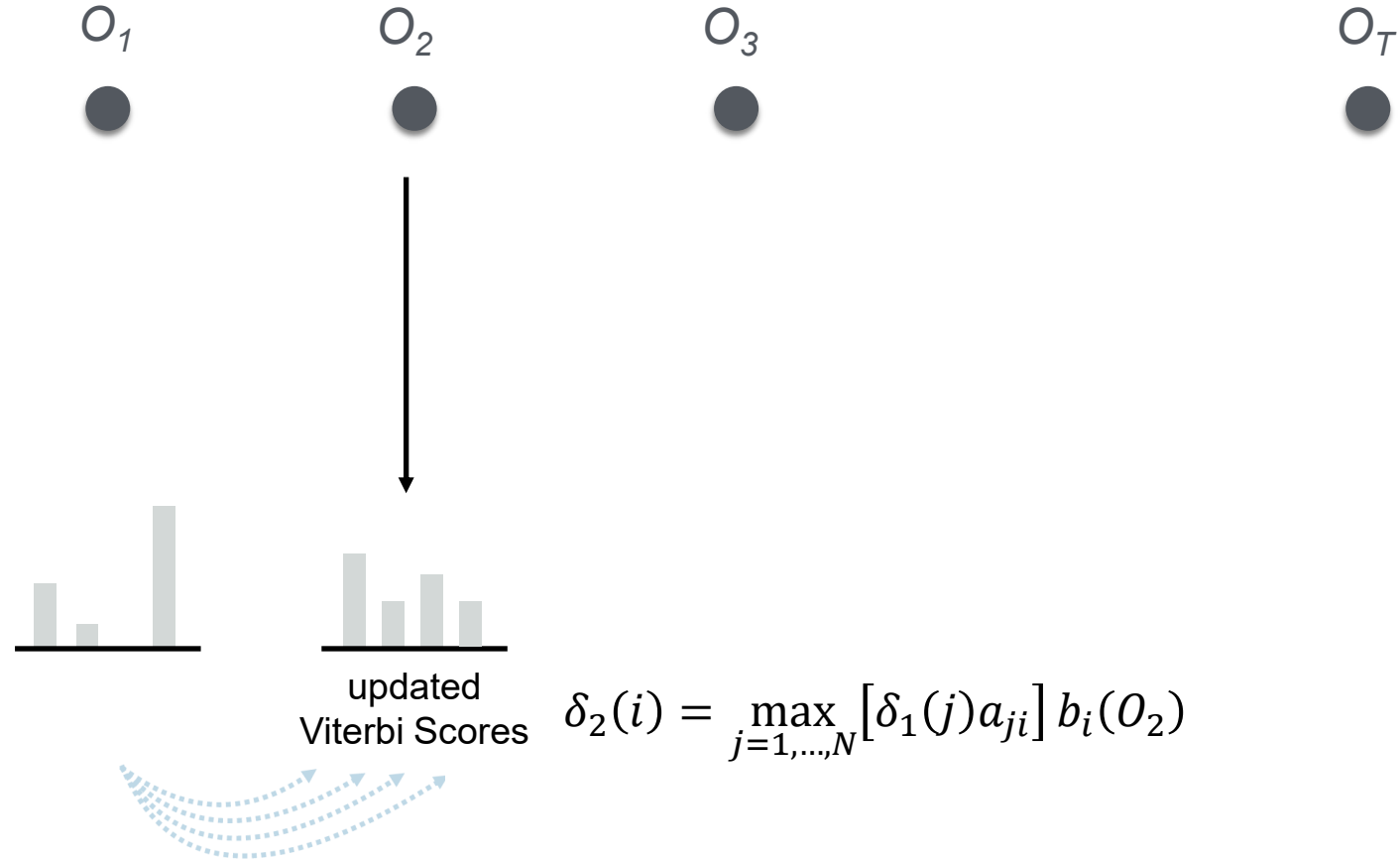


$$\psi_2(i) = \arg \max_{j=1, \dots, N} [\delta_1(j) a_{ji}]$$



Best predecessor
for each state.

2. Calc most likely state sequence



2. Calc most likely state sequence

O_1



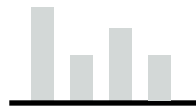
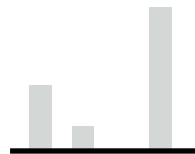
O_2



O_3



O_T

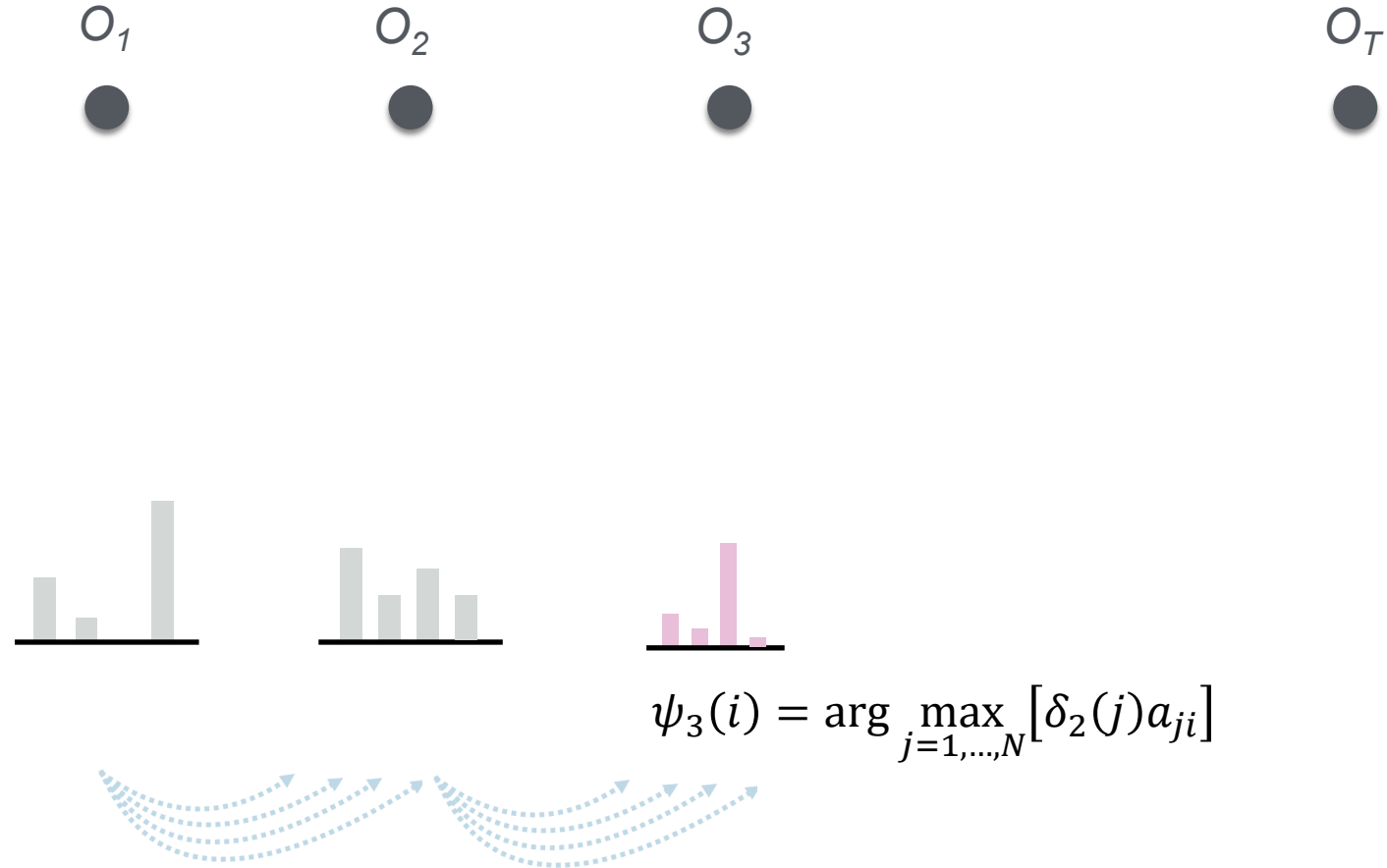


best incoming
scores.

$$\max_{j=1,\dots,N} [\delta_2(j) a_{ji}]$$

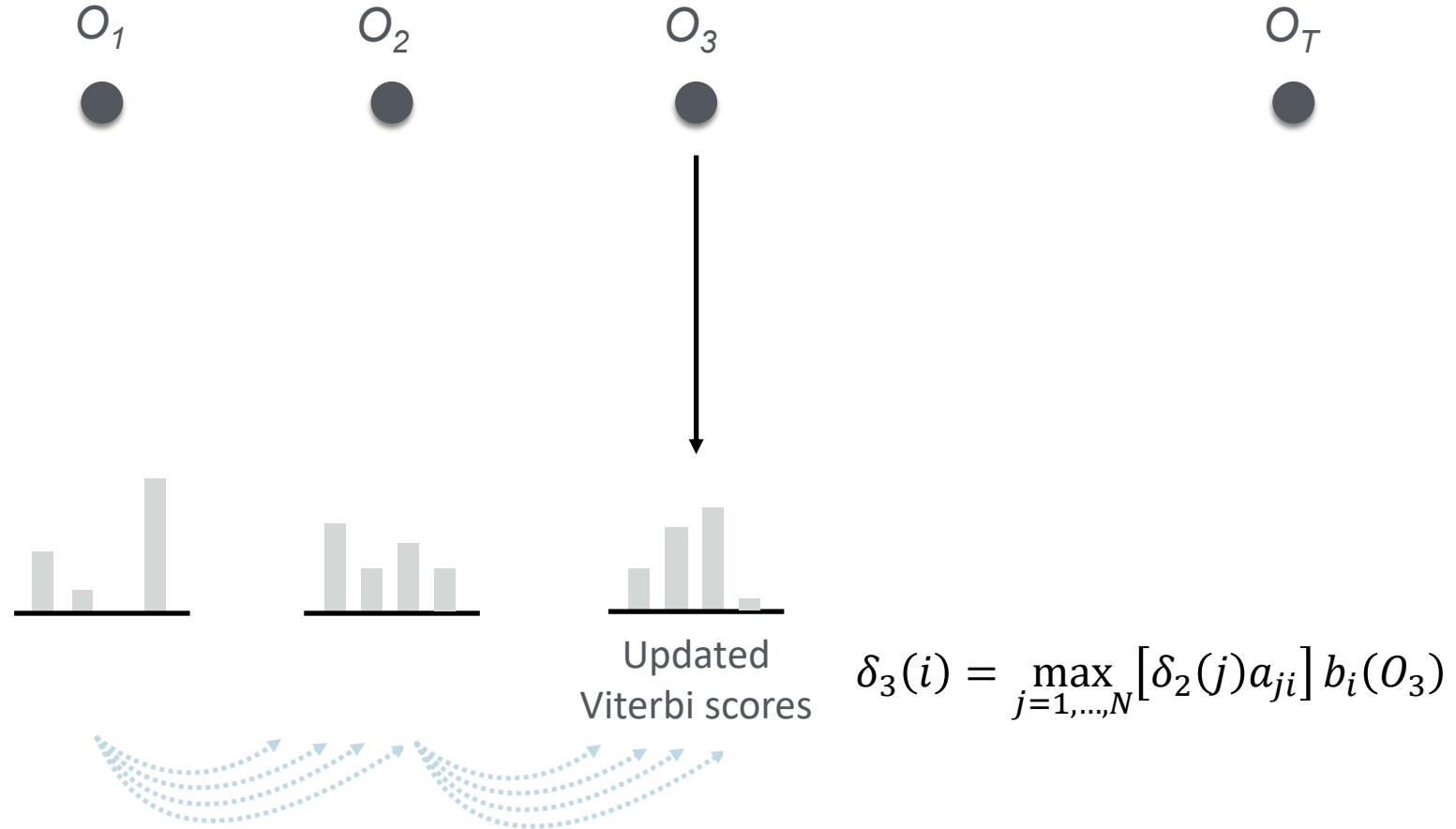


2. Calc most likely state sequence

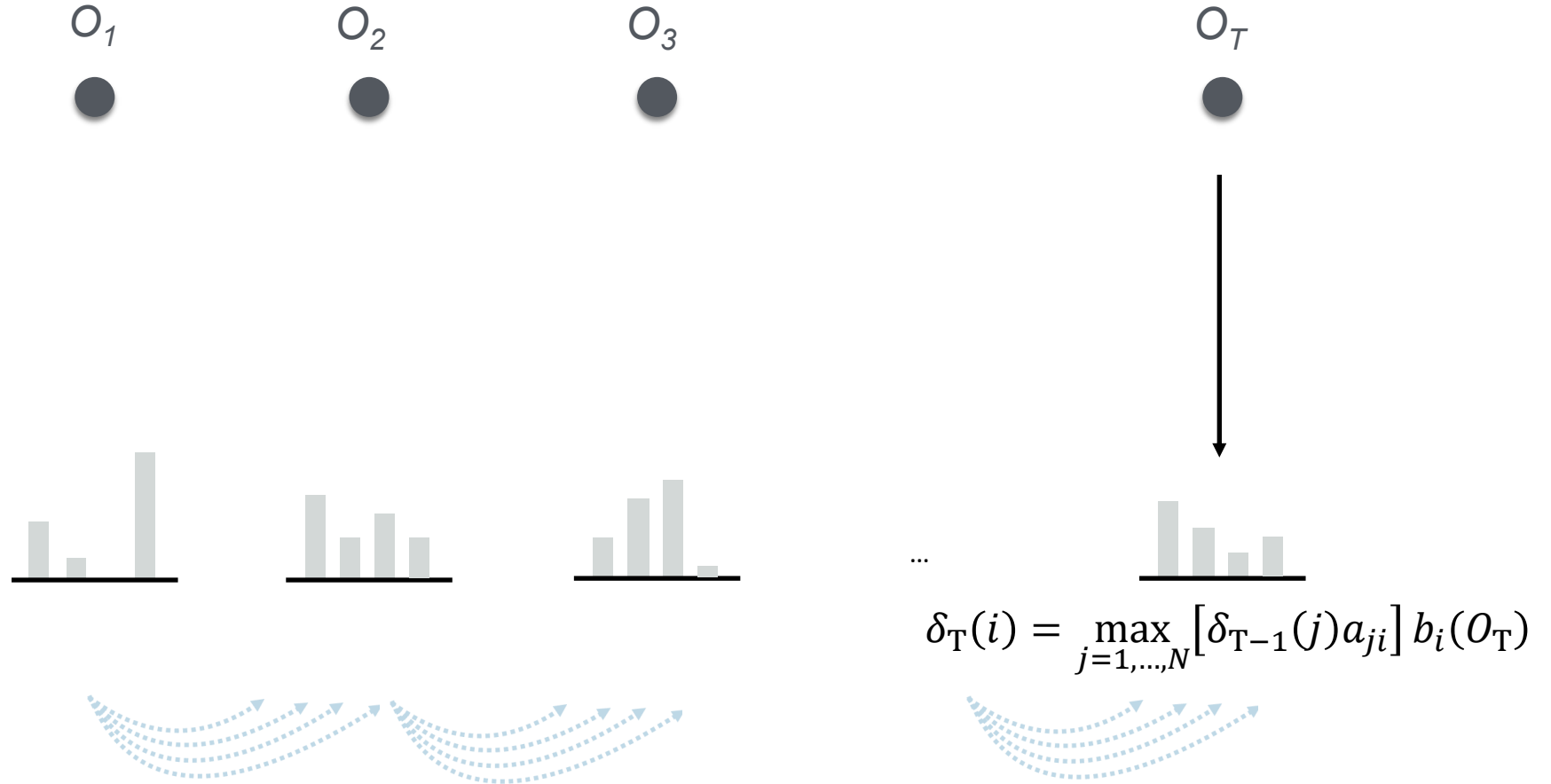


Best predecessor
for each state.

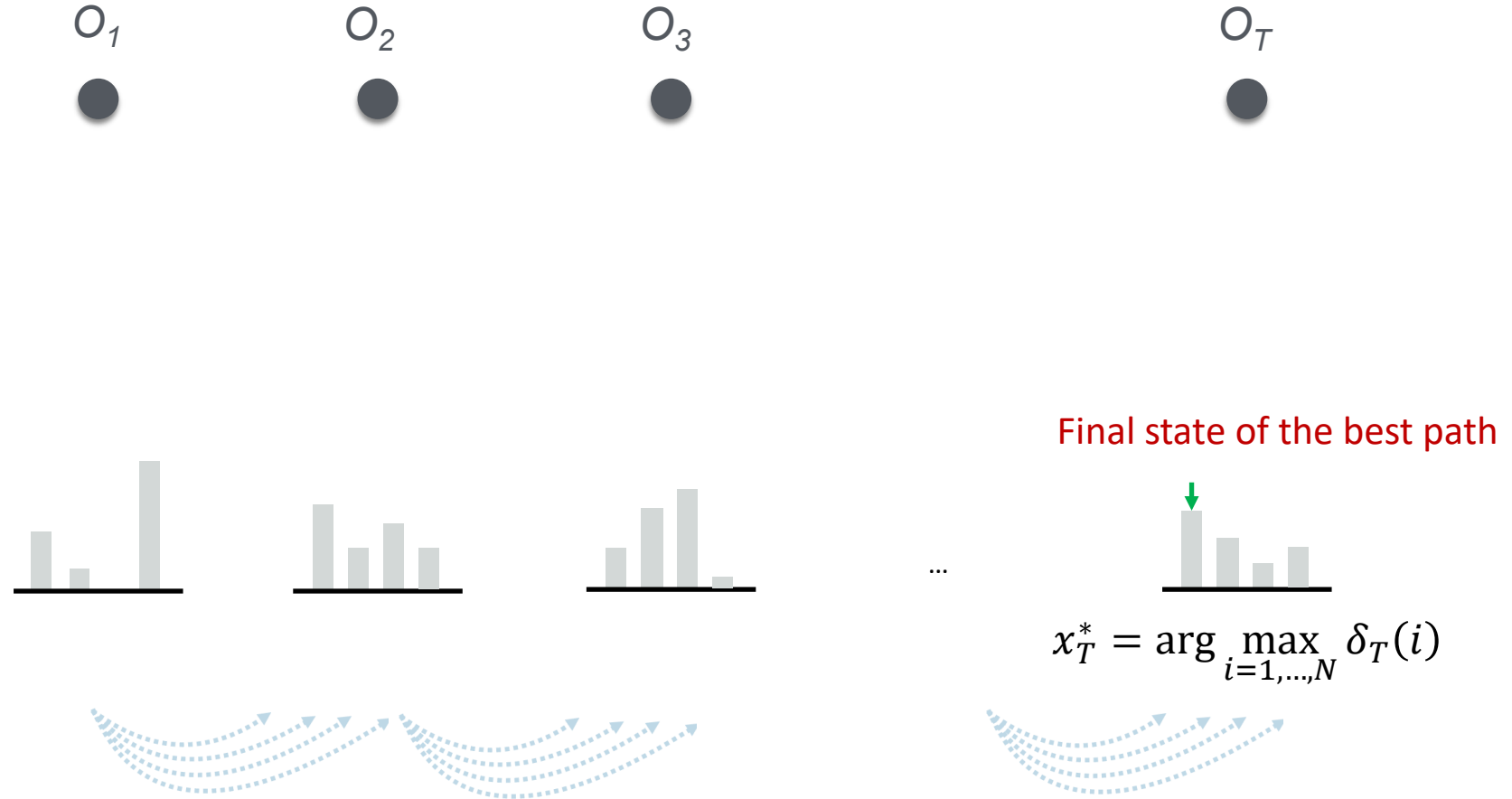
2. Calc most likely state sequence



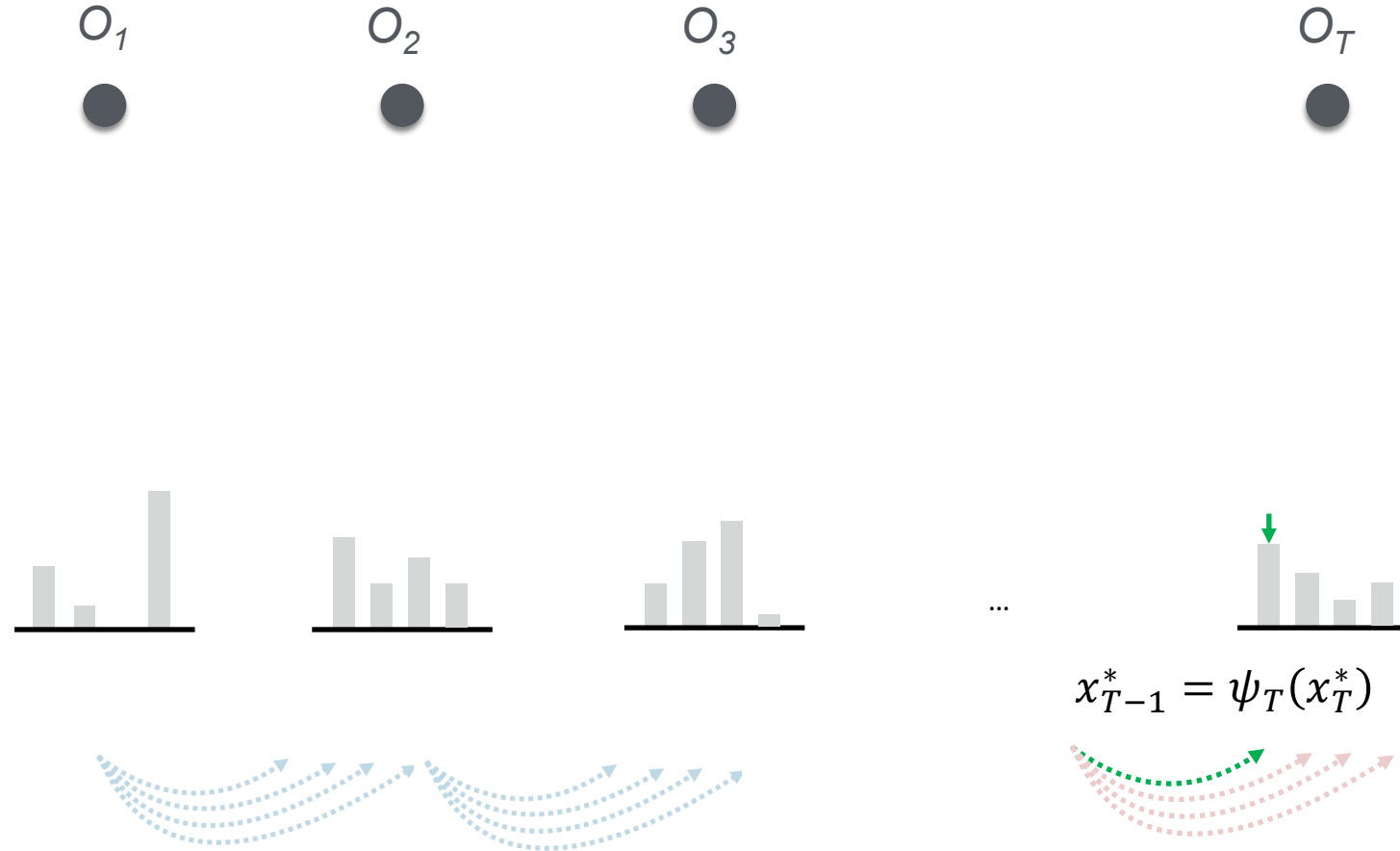
2. Calc most likely state sequence



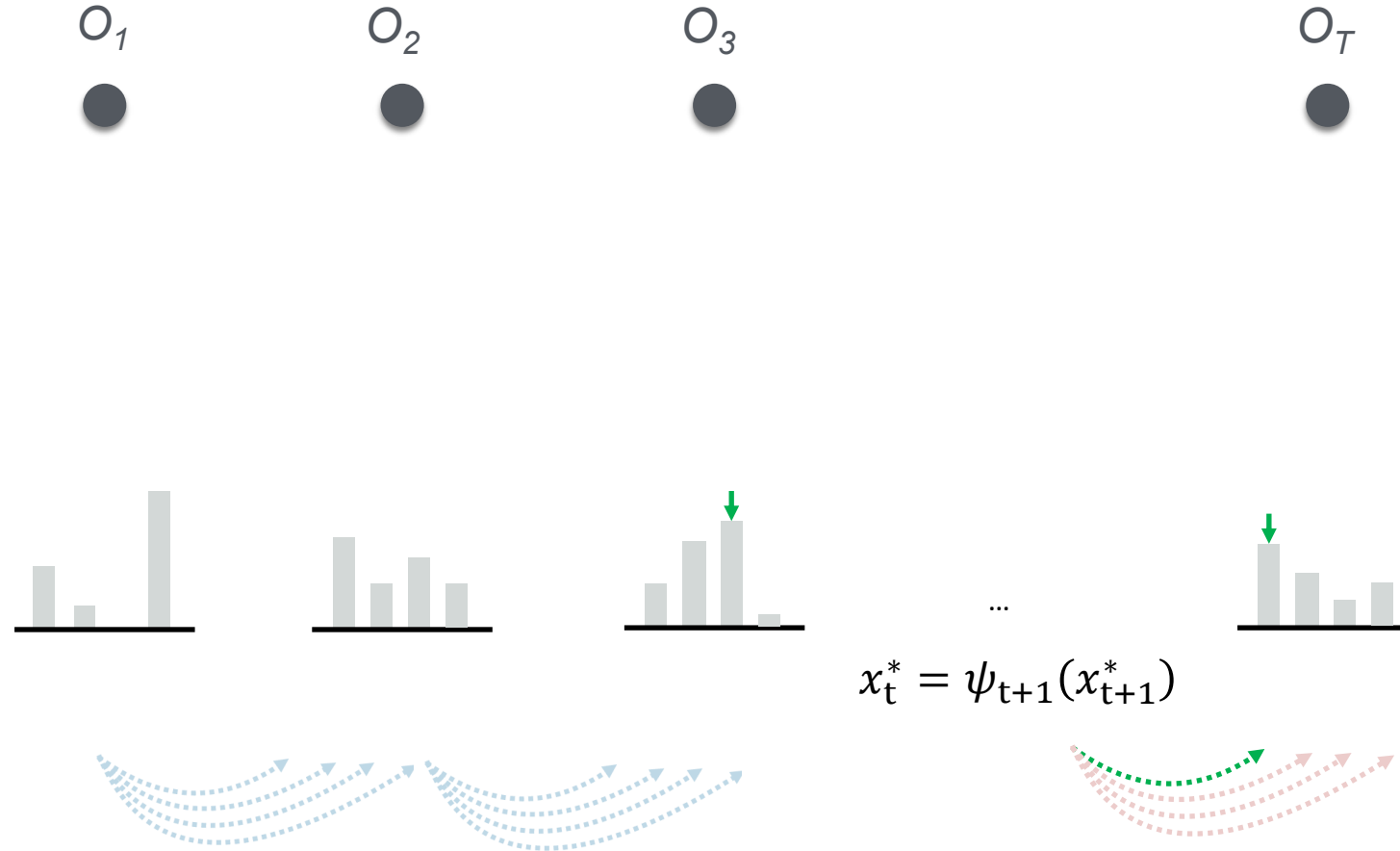
2. Calc most likely state sequence



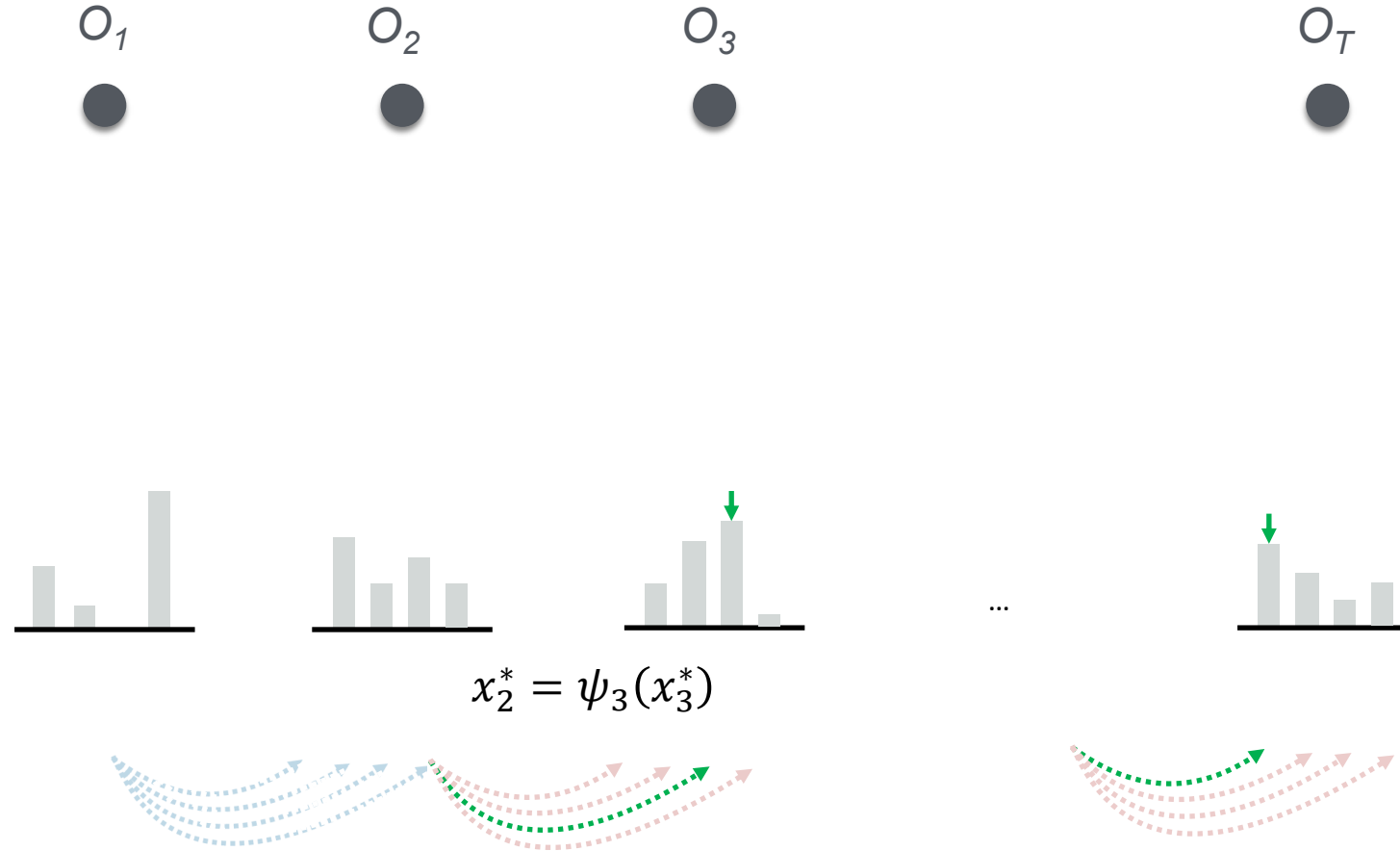
2. Calc most likely state sequence



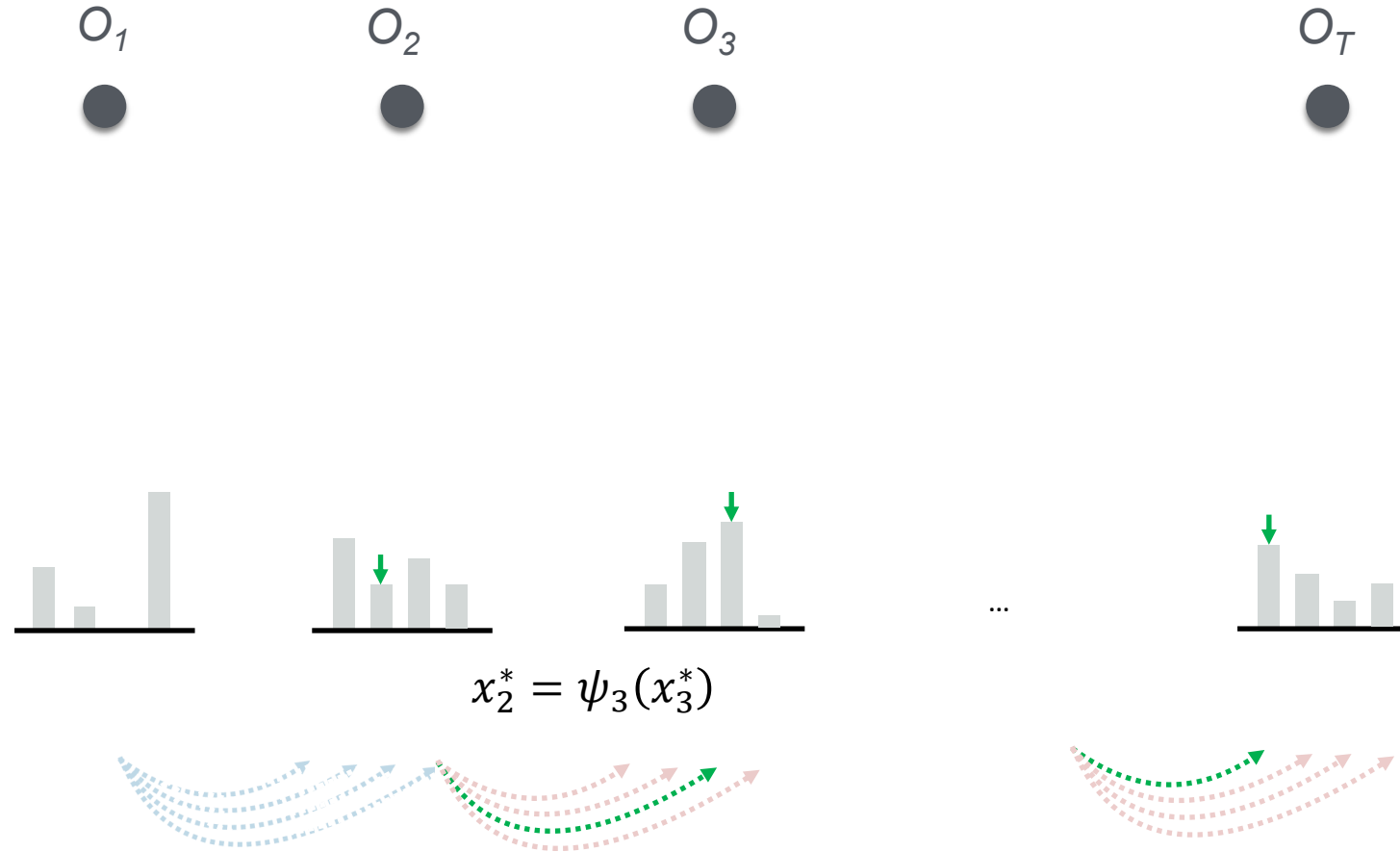
2. Calc most likely state sequence



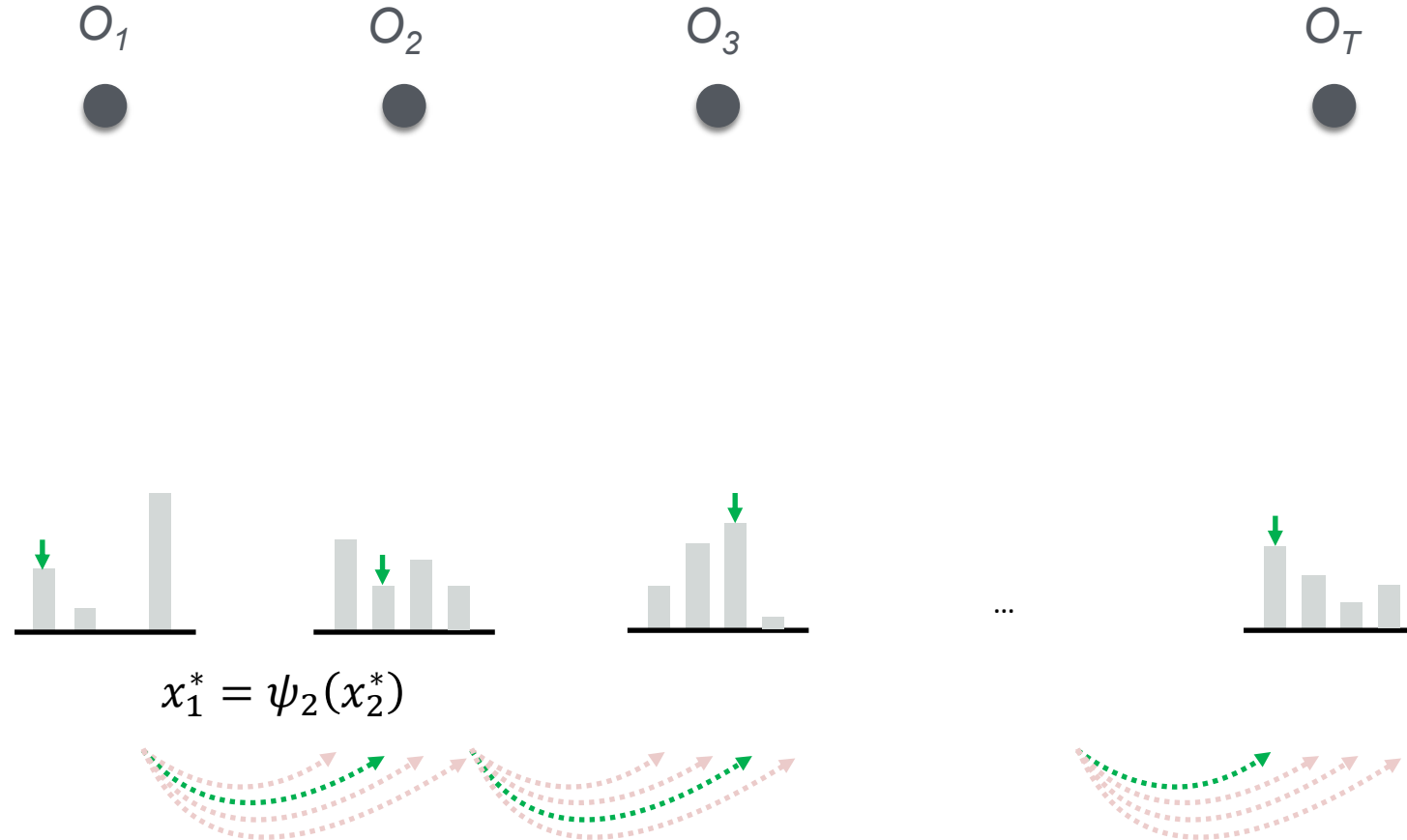
2. Calc most likely state sequence



2. Calc most likely state sequence



2. Calc most likely state sequence



Most likely sequence (Viterbi alg.)

- Solved using dynamic programming (DP) with an algorithm called Viterbi.

- Initialize:

$$\delta_1(i) = \pi_i b_i(O_1), i = 1, \dots, N$$

- For each $t > 1$:

$$\delta_t(i) = \max_{j \in \{1, \dots, N\}} [\delta_{t-1}(j) a_{ji}] b_i(O_t)$$

Partial prob of one of the
most probable paths to
state i at time t

- Probability of best path:

$$\max_{j \in \{1, \dots, N\}} [\delta_T(j)]$$

- Find path by keeping book of preceding states and trace back from highest-scoring final state : $\Psi_{t(i)} = \underset{j \in \{1, \dots, N\}}{\operatorname{argmax}} [\delta_{t-1}(j) a_{ji}]$

Viterbi cont'd

- Watch out for underflows!!! Multiplying many small numbers (probabilities)

- Better to use

- Initialize:

$$\delta_1(i) = \log[\pi_i b_i(O_1)], i = 1, \dots, N$$

- For $t > 1$:

$$\widehat{\delta}_t(i) = \max_{j \in \{1, \dots, N\}} \left[\widehat{\delta}_{t-1}(j) + \log[a_{ji}] + \log[b_i(O_t)] \right]$$

- Log joint probability of best path:

$$\max_{j \in \{1, \dots, N\}} [\widehat{\delta}_T(j)]$$

Three problems solved with HMMs

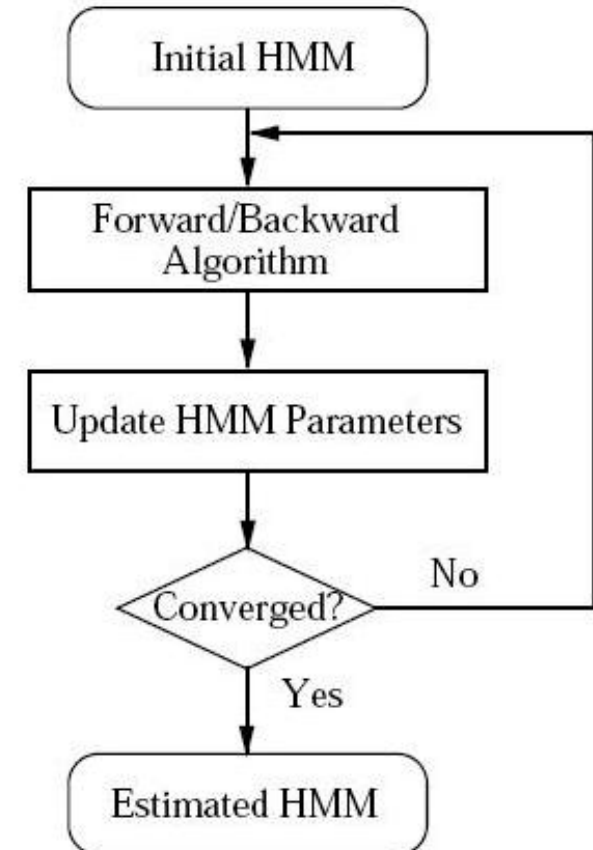
- 1. Evaluation:** Compute likelihood $P(O_{1:t}|\lambda)$
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 $\rightarrow$ A and B matrices and π
Baum-Welch algorithm

3. Estimate model parameters $\Lambda=\{\lambda\}$ given $O_{1:T}$

- Analogous to the training phase of Machine Learning
- Motivating examples:
 - Learn a model for a letter for recognizing handwritten text
 - Learn a model for the word “Bayesian” from audio data
 - Learn a model for the movement of fish in the Fishing Derby Game

The Baum-Welch Algorithm (EM for HMMS)

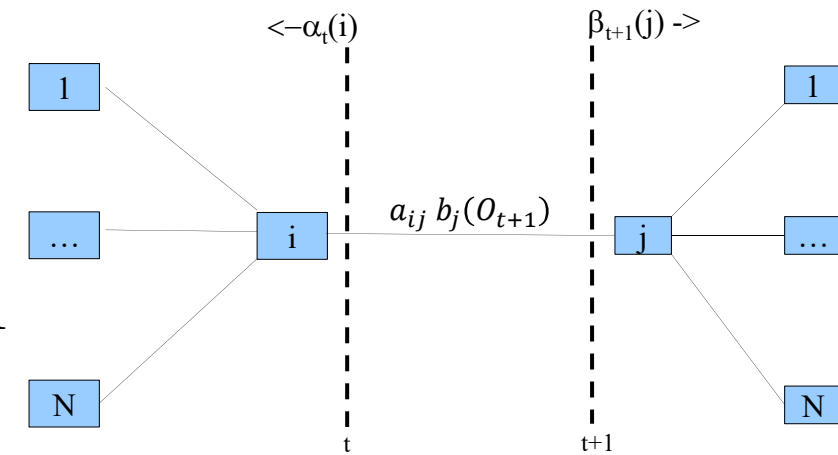
- Given an observation sequence $O_{1:T}$, the number of states, N , and the number of observation outcomes, M .
1. Initialize $\lambda=(A,B,\pi)$
 2. Compute $\alpha_t(i)$, $\beta_t(k)$, $\gamma_t(i,j)$ and $\gamma_t(i)$
 3. Re-estimate the model $\lambda=(A,B,\pi)$
 4. Repeat from 2 until $p(O|\lambda)$ converges



GAMMA CALCULATIONS:

1) Di – Gamma Function

$$\gamma_t(i, j) = \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{r=1}^N \alpha_T(r)} = P(X_t = i, X_{t+1} = j \mid O_{1:T}, \lambda), t = 1, \dots, T - 1$$

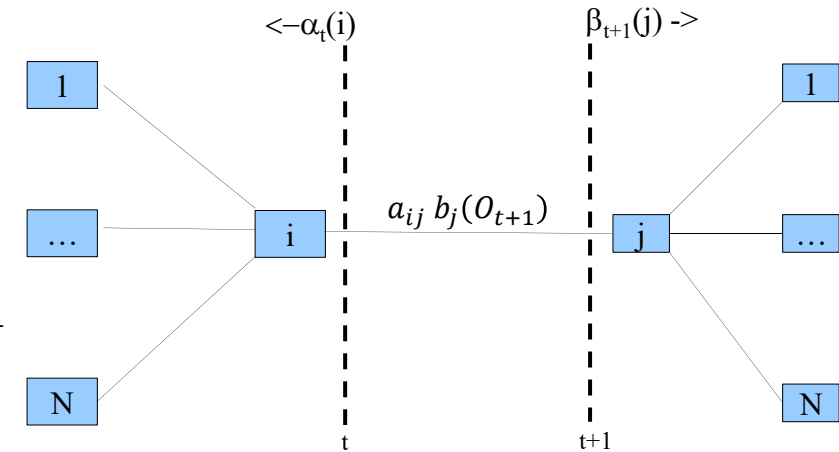


Interpretation: Given the entire observation sequence and current estimate of the HMM, what is the probability that at time (t) the hidden state is ($X_t=i$) & at time (t+1) the hidden state is ($X_{t+1}=j$)?

GAMMA CALCULATIONS:

1) Di – Gamma Function

$$\gamma_t(i, j) = \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{r=1}^N \alpha_T(r)} = P(X_t = i, X_{t+1} = j \mid O_{1:T}, \lambda), t = 1, \dots, T - 1$$



Interpretation: Given the entire observation sequence and current estimate of the HMM, what is the probability that at time (t) the hidden state is ($X_t=i$) & at time (t+1) the hidden state is ($X_{t+1}=j$)?

2) Gamma Function (Marginalizing out X_{t+1})

$$\gamma_t(i) = \sum_{j=1}^N \gamma_t(i, j) = P(X_t = i \mid O_{1:T}, \lambda), t = 1, \dots, T - 1$$

$$\gamma_T(i) = \frac{\alpha_T(i)}{\sum_{r=1}^N \alpha_T(r)}$$

Interpretation: Given the observation sequence and current estimate of the HMM, what is the probability that at time (t) the hidden state is ($X_t=i$)?

Learn the model, i.e. calculating A, B, π

- Given γ and di-gamma, estimate A, B, π using:

A) Transition estimates

$$a_{ij} = \frac{\sum_{t=1}^{T-1} \gamma_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \quad i, j = 1, \dots, N$$

expected number of transitions from state i to state j

expected number of transitions out of state i

B) Emission estimates

$$b_i(k) = \frac{\sum_{t \in \{1, \dots, T\}, o_t = k} \gamma_t(i)}{\sum_{t=1}^T \gamma_t(i)}, \quad i = 1, \dots, N, \quad k = 1, \dots, K$$

expected number of times in state i and observing k

expected number of times in state i , over all T time steps

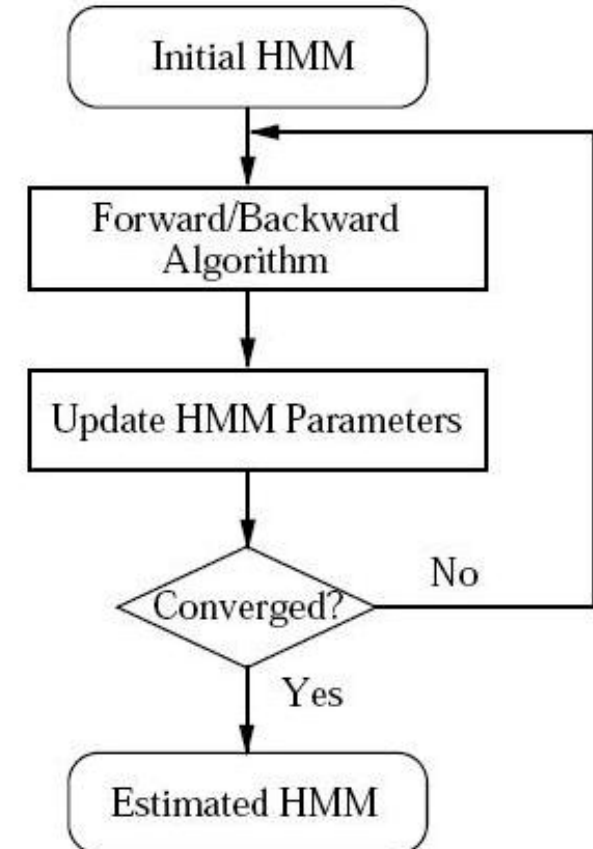
C) Initial state probabilities

$$\pi_i = \gamma_1(i), \quad i = 1, \dots, N$$

expected frequency of being in state i at $t = 1$

The Baum-Welch Algorithm

- Given an observation sequence $O_{1:T}$, the number of states, N , and the number of observation outcomes, M .
1. Initialize $\lambda=(A,B,\pi)$
 2. Compute $\alpha_t(i)$, $\beta_t(i)$, $\gamma_t(i,j)$ and $\xi_t(i,j)$
 3. Re-estimate the model $\lambda=(A,B,\pi)$
 4. Repeat from 2 until $p(O|\lambda)$ converges



To keep in mind

- Baum-Welch is an Expectation-Maximization (EM) algorithm used to train HMM parameters. It *uses* the forward-backward algorithm during each iteration.
- The forward-backward algorithm is just a combination of the forward and backward algorithms: one forward pass, one backward pass.
- With fixed model parameters, forward computes the observation-sequence likelihood, and forward–backward computes posterior probabilities of hidden states. Baum–Welch uses these results to update the parameters.

Model initialization and considerations

- A, B and π must contain valid probabilities: non-negative entries, with each row summing to 1.
- Use prior knowledge for initialization; otherwise, use positive random values and normalize each row.
- Avoid fully symmetric initialization: hidden states may remain indistinguishable during learning.
- Exact-zero probabilities can remain zero under Baum–Welch. Use them deliberately.
- Stop when the improvement in log-likelihood is sufficiently small, or a maximum iteration count is reached.
- Calculating long products of probabilities
 - very small numbers
 - underflow problems
 - use scaling and log likelihoods

Three problems solved with HMMs

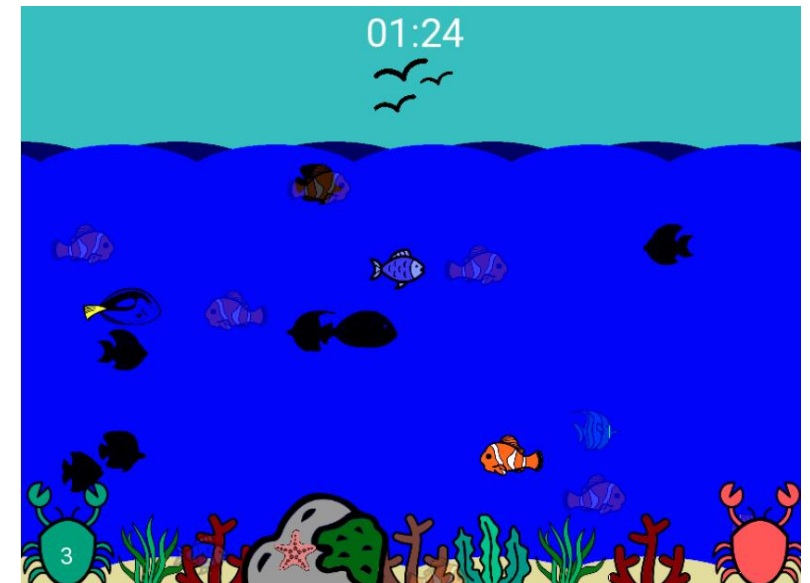
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 $\rightarrow$ A and B matrices and π
Baum-Welch algorithm

Additional (highly recommended) study material

- <http://learn-ai.web.app/>
 - Page developed by one of the previous course TAs with detailed (and visually appealing) explanations of today's lecture exercises + Forward algorithm
 - Visit the page before the HMM Tutorial sessions to consolidate
- Stamp Tutorial on Canvas containing the algorithmic implementations of these algorithms

What's NEXT?

- HMM Tutorials
 - Check the schedule on Canvas and do not forget to attend on Monday, the 14th of September
 - Prepare tutorial exercise sheets (print or .doc)
- Lecture on Multi Agent Systems + Robotics
 - Complete the canvas quiz that will open after the lecture
- Classroom quizzes
 - Sign up for a slot in Canvas and attend on Tuesday, 22nd of September
- HMM assignment will be up after this lecture at 12:00pm!



End of Probabilistic Reasoning Part 2/2 - HMMs

